
A Two-Sided Bayes-Error Bracket from Partition-Conditional Entropy

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Abstract

Many machine-learning predictors are constant on the cells of a fixed partition of their input space: decision trees on leaves, vector quantisers on Voronoi cells, message-passing graph neural networks (MPNNs) on Weisfeiler–Leman colour classes. For every such family the partition is the irreducible expressivity bottleneck. Starting from the classical [Fano \(1961\)](#) and [Hellman & Raviv \(1970\)](#) inequalities, we package a two-sided closed-form bracket between the Bayes error ε_Π^* of the best constant-on-cells predictor of a binary label f and the partition-conditional entropy $H(f \mid \Pi)$,

$$H_{\text{bin}}^{-1}(H(f \mid \Pi)) \leq \varepsilon_\Pi^* \leq \frac{1}{2} H(f \mid \Pi),$$

tight on both boundaries, with maximal ε -width $w^* = \frac{1}{2} H_{\text{bin}}(1/5) - 1/5 \approx 0.1610$ across the entire binary achievable region.

The bracket is elementary; our contribution is what it buys for graph learning. (i) Substituting the depth- L 1-WL partition turns the qualitative Xu–Morris ceiling (“no MPNN exceeds 1-WL”) into a *quantitative* one: no depth- L MPNN can drive transductive training error below $H_{\text{bin}}^{-1}(H(f \mid \Pi_L^{\text{WL}}))$, and a training-free per-cell majority head already attains $\frac{1}{2} H(f \mid \Pi_L^{\text{WL}})$. (ii) We give an *exact* three-term decomposition (Theorem 26) of any trained MPNN’s risk against this ceiling, separating structural ceiling, feature refinement, and head/optimisation slack — three confounds the standard “accuracy after training” comparison entangles. (iii) We prove an ε -robust constancy lemma (Theorem 22) that survives positional encodings and bounded random features, together with a spectral refinement (Theorem 23) replacing the worst-case degree Δ by the adjacency Perron root $\lambda_{\max}(A) \in [\bar{d}, \Delta]$, recovering the empirically observed degree-independence of mean/sym-norm aggregators.

Empirically (§ 9.5): on coarse-partition feature-rich graphs (CiteSeer, PubMed) the WL bracket genuinely bounds realised error; on a heterogeneous architecture menu over UCI Adult it is a statistically significant training-free NAS pre-filter (Kendall $\tau = 0.48$, $p = 5 \times 10^{-5}$) while parameter count *anti*-ranks the same menu. We report the bracket’s failure regimes — partition-cardinality collapse on near-discrete graphs, small- n overfitting on the NAS pre-filter — with equal prominence, and locate both inside the finite-sample theorem (Theorem 28) that also gives the tool its bite on large data. Synthetic verification (`verify.jl`, 10^3 partitions, exact rational + interval arithmetic) closes the loop on the bracket itself; mechanisation of Theorem 4 and Theorem 6 in Lean 4 is in preparation. **Scope.** Finite V , finite Π , binary f .

1 Introduction

From inputs to vertices. Pick any supervised learning problem with a finite or finitely representable input set: a training corpus of n examples, the nodes of a graph, the pixels of an image grid, the leaves of a decision tree, the codewords of a quantiser, the time-steps of a discretised process. Call the elements of this set *vertices* and write the set V . A binary classification task on V is then nothing but a function $f : V \rightarrow \{0, 1\}$ — the *label vector* we wish to predict. This is the most elementary supervised-learning datum: “which vertices are positive?”

What a model actually sees. A learning algorithm rarely processes V directly. Instead, the architecture — a tree depth, a codebook size, a network depth, a feature map — imposes an *equivalence relation* on V : two vertices look identical to the model if and only if they route to the same leaf, the same codeword, the same Weisfeiler–Leman colour, or the same feature-map output. The quotient of V by this equivalence relation is a *partition* $\Pi = \{C_1, \dots, C_m\}$ of V into nonempty *cells* $C_i \subset V$. The architecture-imposed map $C : V \rightarrow \Pi$, sending each vertex to its cell, is the *lens* through which the model perceives the input. Anything the model predicts about a vertex v must factor as

$$\hat{f}(v) = g(C(v)), \quad g : \Pi \rightarrow \{0, 1\}.$$

This factorisation is exact, not approximate: it is forced by the architecture before training begins. In particular, two vertices in the same cell must receive the same predicted label, *regardless of how much training data, parameters, or epochs are spent*. The partition Π is therefore the **irreducible expressivity bottleneck** of the model family.

Three concrete instances. The vertex–partition view unifies models that look superficially unrelated:

- **Decision trees and random forests.** V is the training set; cells C_i are the leaves; the equivalence “ $v \sim w$ iff v and w end at the same leaf” is the partition Π_T . Every leaf-constant prediction rule of CART, C4.5 or any greedy tree learner factors through Π_T (Breiman et al., 1984; Devroye et al., 1996).
- **Vector quantisers and clustering classifiers.** V is the input space (or a finite training sample); cells are pre-images of the k codewords (Voronoi cells in the metric case). Nearest-codeword classification, k -means classifiers and product-quantisation retrieval all factor through their codebook partition Π_Q (Gersho & Gray, 1991; Jégou et al., 2011).
- **Message-passing graph neural networks (MPNNs).** V is the vertex set of a graph G ; the depth- L Weisfeiler–Leman colour refinement $WL_L(G)$ is a partition of V . Every depth- L MPNN with shared, permutation-invariant updates and no node identifiers is constant on the cells of $WL_L(G)$ (Xu et al., 2019; Morris et al., 2019): WL is the lens through which a bounded-depth MPNN sees its graph.

In each case, the architectural choice (tree depth, codebook size, MPNN depth) selects a partition; the partition determines what the model can possibly express; the label f is the only thing the model is asked to fit. The triple (V, Π, f) is the data of the problem.

The two questions. Once one accepts this lens, the practical questions of supervised learning collapse to two:

- (Q1) *Lower bound — no predictor can do better.* Given a partition Π of V and a label $f : V \rightarrow \{0, 1\}$, what is the smallest possible error achievable by *any* function $g : \Pi \rightarrow \{0, 1\}$? We call this the *partition-restricted minimum (empirical) risk* ε_Π^* , deliberately avoiding the term “Bayes error” which in statistical learning theory denotes the population infimum over *all* measurable classifiers. Here ε_Π^* is the minimum risk on V over the architecture-restricted hypothesis class $\{g \circ C : g : \Pi \rightarrow \{0, 1\}\}$, i.e. an empirical quantity under a structural constraint.
- (Q2) *Upper bound — some predictor achieves it.* Is there a quantity, easily computed from (Π, f) , that *certifies* a small ε_Π^* from above? Such a certificate would let a practitioner read off a guaranteed error level without ever training a model.

The question a node-classification practitioner actually asks. The Weisfeiler–Leman ceiling of Xu et al. (2019) and Morris et al. (2019) is the most-cited structural result in graph learning, but it is a *binary verdict*: a pair of graphs is either 1-WL-distinguishable or it is not. That is the right abstraction for graph-classification expressivity, and it is silent on the question a node-classification practitioner faces before

training: *on my graph, with my label vector, how much can a depth- L MPNN possibly recover?* Substituting the depth- L 1-WL partition Π_L^{WL} into the bracket answers this in two numbers per (graph, L) pair — a Fano floor no depth- L MPNN can break, and a constructive ceiling a training-free majority head attains — computed at the cost of the same 1-WL hash that already underlies every GIN implementation. Two consequences organise the rest of the paper. First, the bracket converts the Xu–Morris ceiling from “cannot distinguish” to “cannot drive training error below $H_{\text{bin}}^{-1}(H(f | \Pi_L^{\text{WL}}))$ ”, with the $w^* \approx 0.16$ slack of Theorem 6 as the exact, closed-form-irreducible gap between that floor and the plug-in ceiling. Second — and this is the diagnostic we most want to advance — the bracket *decomposes* the gap between a trained MPNN and its WL ceiling into a structural term (WL-side), a feature-refinement term, and a head/optimisation term (Theorem 26, § 9.3.1). The current convention of comparing architectures by post-training accuracy entangles all three; the bracket separates the first cleanly from the other two, at no training cost on the structural term.

Why entropy? The natural functional of a label $f : V \rightarrow \{0, 1\}$ relative to a partition Π is the *partition-conditional Shannon entropy*

$$H(f | \Pi) := \sum_{C \in \Pi} q_C \cdot H_{\text{bin}}(P_C),$$

where q_C is the mass of cell C and P_C is the fraction of positives inside C . The quantity $H(f | \Pi)$ measures the average *impurity* of the label within cells: it is zero exactly when every cell is monochromatic ($P_C \in \{0, 1\}$ for every C), and reaches 1 exactly when every cell is maximally mixed ($P_C = 1/2$ for every C). Three reasons make $H(f | \Pi)$ inevitable as the right diagnostic:

- *Decision trees already compute it.* CART’s information-gain splitting criterion is the change in $H(f | \Pi)$ when one refines Π at a candidate split. The bracket below converts that internal criterion into an external error guarantee.
- *It is sub- σ -algebra entropy.* In the measure-theoretic view, Π generates a sub- σ -algebra of the power set of V , and $H(f | \Pi)$ is the standard conditional entropy of the random label f given this sub- σ -algebra (Cover & Thomas, 2006, §2.1). Every partition-respecting predictor is measurable with respect to this sub- σ -algebra; conditional entropy is the canonical information-theoretic obstruction to recovering f from such predictors.
- *It is the only functional with a two-sided bracket.* Classical inequalities of Fano (1961) and Hellman & Raviv (1970) relate ε_{Π}^* to $H(f | \Pi)$ from both sides; no other elementary scalar functional of (Π, f) enjoys both a matching lower bound *and* a matching upper bound in closed form.

This paper. For binary labels we answer (Q1) and (Q2) simultaneously by a single classical inequality recast for partitions: the partition-conditional Shannon entropy $H(f | \Pi)$ two-sidedly brackets ε_{Π}^* :

$$H_{\text{bin}}^{-1}(H(f | \Pi)) \leq \varepsilon_{\Pi}^* \leq \frac{1}{2} H(f | \Pi).$$

The lower side specialises Fano’s inequality (Fano, 1961); the upper side specialises the Hellman–Raviv bound (Hellman & Raviv, 1970); the two-sided packaging is in the spirit of Feder & Merhav (1994) and the modern treatment of Ho & Verdú (2010). The contribution of this paper is not the inequalities themselves but their packaging as a *single architecture-level diagnostic* on (V, Π, f) : once computed, the bracket gives the practitioner *simultaneously* an unconditional lower bound on the error of every model in the architectural class, and a constructive upper bound achieved by the elementary plug-in predictor on cells.

What the bracket enables. A practitioner armed with $H(f | \Pi)$ for a candidate architecture can:

- (E1) *Reject architectures cheaply (the underfitting floor).* If $H_{\text{bin}}^{-1}(H(f | \Pi))$ exceeds the target error on V , no training run of any model in the family can fit the data below that target; reject the family before spending compute. This is a *one-sided* diagnostic: a high empirical-risk floor is necessarily inherited by any population risk (training cannot beat its own ceiling).

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- (E2) *Certify architectural capacity cheaply.* If $\frac{1}{2}H(f \mid \Pi)$ is smaller than the target error, the plug-in predictor on cells already meets the target *on* V , with no optimiser, no validation set, and no hyperparameters. This certifies that the architecture has enough expressive capacity on V ; it does *not* by itself guarantee out-of-sample generalisation, which requires the population extension of Theorem 28.
 - (E3) *Compare architectures on equal footing.* Two different partitions Π_1, Π_2 (say, tree depths or MPNN depths) yield two brackets; the one with smaller $H(f \mid \Pi)$ dominates uniformly in the optimiser.
 - (E4) *Track refinement gains.* If Π' refines Π (deeper tree, larger codebook, more MPNN layers), $H(f \mid \Pi') \leq H(f \mid \Pi)$ and both bracket endpoints move monotonically (Theorem 12). This quantifies the marginal value of model-class growth.

Cost and scope. “Training-free” here means *optimisation-free*: computing the bracket requires the labels f (so the diagnostic is supervised, not unsupervised), but it requires no gradient steps, parameter initialisation, learning-rate tuning, or validation set. The cost of evaluating $H(f \mid \Pi)$ is a single $O(|V|)$ pass to accumulate cell frequencies P_C and masses q_C ; this is strictly dominated by the cost of one training epoch. On four binary tabular benchmarks ranging from $n = 178$ to $n = 45,222$ we measure (Table 15) that the bracket is between $20\times$ and $133\times$ cheaper than fitting a single CART tree of the same cell budget, and cheaper than one LBFGS epoch of logistic regression on three of the four datasets.

The bracket is tight on both boundaries by explicit witnesses, and we show in Theorem 14 that no closed-form improvement is possible. The bracket width is bounded above by an explicit constant $w^* \approx 0.1610$ uniformly in Π and f — the maximal ε -extent of the binary achievable region of Feder & Merhav (1994), whose closed form $\frac{1}{2}H_{\text{bin}}(1/5) - 1/5$ we record here.¹

Contributions.

- (C1) *Bracket.* A self-contained two-sided closed-form bracket (Theorem 4) on the partition-restricted minimum risk of every partition-constant predictor of a binary label, with an elementary finite-sum proof.
- (C2) *Universal slack constant.* The explicit constant $w^* = \frac{1}{2}H_{\text{bin}}(1/5) - 1/5 \approx 0.1610$ (Theorem 6), bounding the bracket’s width uniformly over *all* finite partitions and *all* binary labels. This is the maximal ε -extent of the binary achievable region of Feder & Merhav (1994); we make no priority claim for the constant itself (a direct calculus consequence of standard inequalities), but record its closed form and deploy it as the uniform conservativeness ceiling driving every downstream application of this paper.
- (C3) *Marginal-aware refinement.* A sharper bracket (Theorem 7) using the global label marginal π : the marginal-aware slack $w^*(\pi_*)$ shrinks strictly below w^* for unbalanced labels, e.g. $w^*(0.1) \approx 0.069$.
- (C4) *Empirical-to-population extension.* A standard concentration upgrade (Theorem 28) converting the bracket on V into a PAC-style population diagnostic with an $O(\sqrt{\log m/n})$ additive correction under bounded cell mass and bounded cell mean.
- (C5) *Sharpness.* A negative result (Theorem 14) ruling out any uniform improvement of either side by simpler closed-form expressions.
- (C6) *Three worked applications.* Decision trees, vector quantisation, and Weisfeiler–Leman-refined message passing (§ 9), each instantiated with numerical examples, yielding training-free error brackets for that family.

¹In the binary case the Fano lower curve and the Hellman–Raviv upper line are *exactly* the boundaries of the achievable region (Theorem 5), so w^* is the genuine maximal width of the realisable (ε, H) set, not merely the gap between two bounds. We make no priority claim for the constant itself, which is a direct calculus consequence of standard inequalities.

(C7) *Mechanised check.* A short Julia script (`verify.jl`, shipped with this preprint) verifies the bracket on 10^3 random partitions in exact rational arithmetic (for ε_{Π}^*) together with certified interval arithmetic (for the Shannon-entropy terms); zero violations.

(C8) *Empirical evidence on real benchmarks.* Eight self-contained experiments (§ 9.5) on UCI Adult, breast-cancer, wine, digits-binary, Twitch-EN, Cora, CiteSeer, PubMed, ogbn-arxiv, and a featureless structural-WL battery (Cayley + Paley). Highlights: on UCI Adult, CART training error matches $\varepsilon_{\Pi_d}^*$ to four decimal places at every depth $d = 1, \dots, 15$ (Table 2) and logistic-regression-on-cells matches $\varepsilon_{\Pi_k}^*$ for every $k \leq 256$ (Table 3); per-arch bracket evaluation is $20\text{--}133\times$ cheaper than one CART fit (Table 15); as a heterogeneous-menu NAS pre-filter the bracket attains Kendall $\tau = 0.48$ ($p = 5 \times 10^{-5}$) with held-out test error on Adult while parameter count *anti*-ranks the same menu ($\tau = -0.38$, Table 14); on feature-rich citation / social graphs the depth-3 1-WL partition becomes near-discrete (95.6% singleton cells on ogbn-arxiv), exposing the bracket’s partition-cardinality collapse regime which we report transparently rather than claim as a victory (Table 5); on featureless structural-WL benchmarks the bracket correctly pins at the marginal-entropy ceiling (Table 6); empirical concentration to the population bracket matches the $\Theta(1/\sqrt{n})$ rate of Theorem 28 with 1.000 coverage at the nominal 0.95 level (Table 16).

Scope. All statements are for finite V , finite partition Π , and binary f . Multi-class extensions ($|\mathcal{Y}| > 2$) proceed via the concave-envelope construction of Feder & Merhav (1994) and are not pursued here. Continuous-alphabet generalisations require additional regularity (Ho & Verdú, 2010) and are also not pursued.

Related work. See § 10 for placement against the entropy-bounds-error literature (Fano, 1961; Hellman & Raviv, 1970; Tebbe & Dwyer III, 1968; Kovalevskij, 1968; Feder & Merhav, 1994; Han & Verdú, 1994; Ho & Verdú, 2010; Sason & Verdú, 2018), the message-passing expressivity literature (Xu et al., 2019; Morris et al., 2019; Böker et al., 2023), and decision-tree / VQ foundations (Breiman et al., 1984; Gershon & Gray, 1991; Devroye et al., 1996).

2 Setup

We now make the vertex-partition-label triple (V, Π, f) of § 1 formal, and fix the two quantities that the rest of the paper relates: the partition-restricted minimum risk ε_{Π}^* (the quantity to be bounded) and the partition-conditional entropy $H(f \mid \Pi)$ (the bound).

Let V be a finite set with $|V| = n \geq 1$, and let Π be a partition of V into nonempty cells $\Pi = \{C_1, \dots, C_m\}$. Let $f : V \rightarrow \{0, 1\}$ be a binary label, and write $C(v)$ for the unique cell of Π containing $v \in V$. Endow V with the uniform probability measure $q_v = 1/n$ (Theorem 2 discusses non-uniform weights).

Cell statistics. For each cell $C \in \Pi$ write

$$q_C := |C|/n, \quad P_C := |C|^{-1} \sum_{v \in C} f(v), \quad e_C := \min(P_C, 1 - P_C) \in [0, 1/2].$$

Thus q_C is the mass of cell C , P_C is the cell-conditional mean of f , and e_C is the within-cell majority-prediction error.

Partition-restricted minimum risk. The minimum error of any constant-on-cells predictor is

$$\varepsilon_{\Pi}^* := \min_{g: \Pi \rightarrow \{0, 1\}} \mathbb{P}[g(C(v)) \neq f(v)] = \sum_{C \in \Pi} q_C \cdot e_C, \quad (1)$$

attained by the plug-in rule $\hat{h}_{\Pi}(v) := \mathbf{1}\{P_{C(v)} \geq 1/2\}$. Equivalently, ε_{Π}^* is the Bayes risk of f when the predictor is restricted to depend on v only through $C(v)$ (Devroye et al., 1996, §2.1).

Partition-conditional entropy. In bits,

$$H(f | \Pi) := \sum_{C \in \Pi} q_C \cdot H_{\text{bin}}(P_C) = \sum_{C \in \Pi} q_C \cdot H_{\text{bin}}(e_C),$$

where $H_{\text{bin}}(p) := -p \log_2 p - (1-p) \log_2 (1-p)$ with $0 \log_2 0 := 0$, and the second equality is the symmetry $H_{\text{bin}}(p) = H_{\text{bin}}(1-p)$.

Inverse binary entropy. The function $H_{\text{bin}} : [0, 1] \rightarrow [0, 1]$ is strictly concave, symmetric about $1/2$, with maximum $H_{\text{bin}}(1/2) = 1$, strictly increasing on $[0, 1/2]$. Its inverse on the increasing branch is the unique

$$H_{\text{bin}}^{-1} : [0, 1] \rightarrow [0, 1/2]$$

satisfying $H_{\text{bin}}(H_{\text{bin}}^{-1}(h)) = h$. As the inverse of a concave increasing function, H_{bin}^{-1} is *convex*, continuous, and strictly increasing, with $H_{\text{bin}}^{-1}(0) = 0$ and $H_{\text{bin}}^{-1}(1) = 1/2$.

Remark 1 (range of $H(f | \Pi)$). Since $H_{\text{bin}}(P_C) \in [0, 1]$ for every C and $\sum_C q_C = 1$, the partition-conditional entropy satisfies $H(f | \Pi) \in [0, 1]$ in bits. In particular, $H(f | \Pi)$ lies in the domain of H_{bin}^{-1} , justifying the left-hand side of (2) below.

Remark 2 (non-uniform weighting). All statements and proofs below extend verbatim to non-uniform weights $\{q_v\}_{v \in V}$ with $\sum_v q_v = 1$, by redefining $q_C := \sum_{v \in C} q_v$ and $P_C := q_C^{-1} \sum_{v \in C} q_v f(v)$. The only properties of $\{q_C\}$ used are nonnegativity and $\sum_C q_C = 1$.

Remark 3 (empirical vs population, notation). Throughout § 3 and ??? the symbol ε_{Π}^* denotes the *empirical* partition-restricted minimum risk on the finite set V under the uniform (or weighted) measure $\{q_v\}$. Its *population* analogue

$$\varepsilon_{\Pi, \mu}^* := \sum_{C \in \Pi} \mu(C) \min(\bar{P}_C, 1 - \bar{P}_C), \quad \bar{P}_C := \mathbb{E}_{\mu}[f | C],$$

defined when (V, Π, f) is an i.i.d. draw from a population distribution μ on a measurable space, is the object of Theorem 28 and concentrates on ε_{Π}^* at the $\mathcal{O}(1/\sqrt{n})$ rate established there. Practitioner claims (E1)–(E3) in ?? bound the empirical ε_{Π}^* unconditionally; their population reading $\varepsilon_{\Pi}^* \approx \varepsilon_{\Pi, \mu}^*$ is mediated by Theorem 28 and inherits its $\kappa(\delta, \eta)$ caveats (in particular the $\eta_{\min} \rightarrow 0$ degeneracy of E7’s near-pure CART leaf).

3 Main result

Theorem 4 (Two-sided bracket). *For every finite partition Π of a finite set V and every binary $f : V \rightarrow \{0, 1\}$,*

$$H_{\text{bin}}^{-1}(H(f | \Pi)) \leq \varepsilon_{\Pi}^* \leq \frac{1}{2} H(f | \Pi). \quad (2)$$

The two inequalities are respectively the lower and upper bounds of Ho & Verdú (2010) specialised to the partition channel, in turn refining classical results of Fano (1961) and Hellman & Raviv (1970) packaged in the achievable-region form of Feder & Merhav (1994). The contribution of this paper is not the theorem itself but its packaging and consequences (Theorems 6 and 14 and § 9).

Achievable region. Define the candidate region

$$\tilde{A}_2 := \{(\varepsilon, H) \in [0, 1/2] \times [0, 1] : H_{\text{bin}}^{-1}(H) \leq \varepsilon \leq H/2\},$$

shown in Figure 1.

Proposition 5 (Achievable region). *$\tilde{A}_2$ is exactly the closure of the set of pairs $(\varepsilon_{\Pi}^*, H(f | \Pi))$ achievable by some finite partition Π of a finite set V and some binary label $f : V \rightarrow \{0, 1\}$. Concretely:*

- (i) *Every achievable pair lies in $\tilde{A}_2$ (Theorem 4).*
- (ii) *The lower boundary $\{(\varepsilon, H_{\text{bin}}(\varepsilon)) : \varepsilon \in [0, 1/2]\}$ is realised by the trivial-partition Fano family $\Pi_{\varepsilon}^{\text{F}}$ of § 8.*

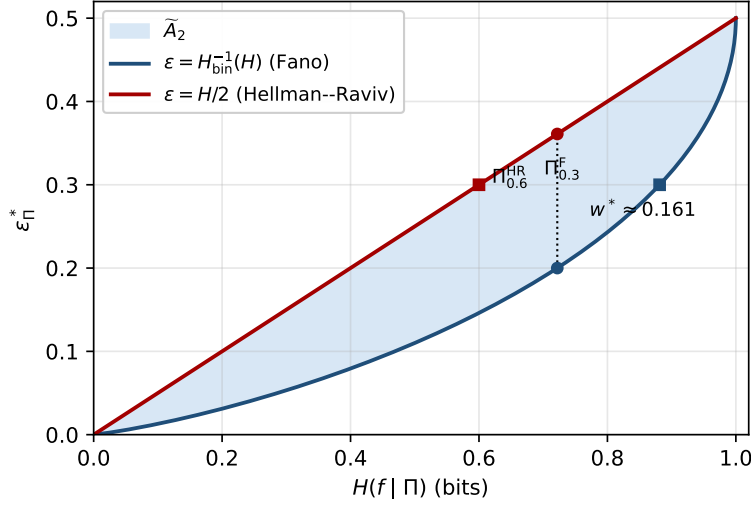


Figure 1: The achievable region $\tilde{A}_2$ on the (H, ϵ) plane. Upper boundary $\epsilon = H/2$ (Hellman–Raviv, tight via Π_α^{HR}). Lower boundary $\epsilon = H_{\text{bin}}^{-1}(H)$ (Fano, tight via Π_ϵ^{F}). Maximum width $w^* \approx 0.1610$ attained at $\epsilon = 1/5$. Every $(\epsilon_\Pi^*, H(f | \Pi))$ from a finite partition and binary label lies in the shaded region; both boundaries and the interior are achievable.

- (iii) The upper boundary $\{(H/2, H) : H \in [0, 1]\}$ is realised by the two-cell Hellman–Raviv family Π_α^{HR} of § 8 with $\alpha = H$.
- (iv) Every interior point of $\tilde{A}_2$ is realised by a disjoint union of one boundary witness with appropriate mass split (Theorem 16).

The proof is collected after the boundary witnesses are defined in § 8; see Theorem 16 for the explicit interior construction.

Corollary 6 (Uniform slack). *Let $w(H) := \frac{1}{2}H - H_{\text{bin}}^{-1}(H)$ be the width of the bracket (2) at conditional entropy H . Then w vanishes at $H \in \{0, 1\}$, is concave on $[0, 1]$, and attains its maximum*

$$w^* := \max_{H \in [0, 1]} w(H) = \frac{1}{2} H_{\text{bin}}(1/5) - 1/5 \approx 0.1610$$

at $H^* = H_{\text{bin}}(1/5) \approx 0.7219$, equivalently at $\epsilon = 1/5$. Consequently, for every Π and f ,

$$\frac{1}{2} H(f | \Pi) - \epsilon_\Pi^* \leq w^* \quad \text{and} \quad \epsilon_\Pi^* - H_{\text{bin}}^{-1}(H(f | \Pi)) \leq w^*.$$

Proof. $w'(H) = \frac{1}{2} - 1/H'_{\text{bin}}(H_{\text{bin}}^{-1}(H))$ by the chain rule. Setting $w'(H) = 0$ gives $H'_{\text{bin}}(\epsilon) = 2$, equivalently $\log_2((1 - \epsilon)/\epsilon) = 2$, i.e. $\epsilon = 1/5$. Substituting back gives $H^* = H_{\text{bin}}(1/5)$ and the value w^* . Concavity of w follows from concavity of $H/2$ (linear) and convexity of H_{bin}^{-1} (so $-H_{\text{bin}}^{-1}$ is concave). Vanishing at the endpoints is immediate. $\square$

4 Marginal-aware refinement

The slack of Theorem 6 is uniform in Π and f , but it is loose when the global label marginal is far from $1/2$: no constant-on-cells predictor can possibly do worse than the trivial *global majority* predictor, and incorporating this trivial upper bound tightens the upper side of (2) for unbalanced labels.

Proposition 7 (Marginal-aware bracket). *Let $\pi := \sum_{v \in V} q_v f(v) \in [0, 1]$ be the global mean of f , where $q_v := 1/n$, and write $\pi_* := \min(\pi, 1 - \pi) \in [0, 1/2]$. Then*

$$H_{\text{bin}}^{-1}(H(f | \Pi)) \leq \epsilon_\Pi^* \leq \min\left(\frac{1}{2}H(f | \Pi), \pi_*\right). \quad (3)$$

The associated marginal-aware slack

$$w^*(\pi_*) := \max_{H \in [0,1]} [\min(\frac{1}{2}H, \pi_*) - H_{\text{bin}}^{-1}(H)]$$

admits the closed form

$$w^*(\pi_*) = \begin{cases} w^* \approx 0.1610, & \pi_* \geq \frac{1}{2}H_{\text{bin}}(1/5) \approx 0.3610, \\ \pi_* - H_{\text{bin}}^{-1}(2\pi_*), & \pi_* < \frac{1}{2}H_{\text{bin}}(1/5), \end{cases} \quad (4)$$

and satisfies $w^*(\pi_*) \leq w^*$ with equality iff $\pi_* \geq \frac{1}{2}H_{\text{bin}}(1/5)$.

Proof. The constant predictor $g \equiv \mathbf{1}\{\pi \geq 1/2\}$ on V has error rate π_* and is trivially constant on every cell, hence $\varepsilon_{\Pi}^* \leq \pi_*$. Combining with Theorem 4 gives (3).

For the slack, let $\phi(H) := \min(\frac{1}{2}H, \pi_*) - H_{\text{bin}}^{-1}(H)$. On $[0, 2\pi_*]$, $\phi(H) = w(H) = \frac{1}{2}H - H_{\text{bin}}^{-1}(H)$, the function of Theorem 6, which is concave on $[0, 1]$ with unique critical point $H^* = H_{\text{bin}}(1/5)$ and maximal value w^* . On $[2\pi_*, 1]$, $\phi(H) = \pi_* - H_{\text{bin}}^{-1}(H)$ is strictly decreasing in H , so the maximum of ϕ over $[2\pi_*, 1]$ is attained at $H = 2\pi_*$ with value $\pi_* - H_{\text{bin}}^{-1}(2\pi_*)$. Two cases:

- If $2\pi_* \geq H^*$, i.e. $\pi_* \geq \frac{1}{2}H_{\text{bin}}(1/5)$, the global maximum of w on $[0, 1]$ lies inside $[0, 2\pi_*]$, so $w^*(\pi_*) = w^*$.
- If $2\pi_* < H^*$, then w is strictly increasing on $[0, 2\pi_*]$ (as H^* is its unique critical point and $w(0) = 0$), so the maximum over $[0, 2\pi_*]$ is at $H = 2\pi_*$ with value $\pi_* - H_{\text{bin}}^{-1}(2\pi_*)$, which also matches the right-branch value, giving $w^*(\pi_*) = \pi_* - H_{\text{bin}}^{-1}(2\pi_*)$.

The inequality $w^*(\pi_*) \leq w^*$ is then immediate. $\square$

Theorem 7 reproduces the universal slack $w^* \approx 0.1610$ at $\pi_* = 1/2$ and shrinks it substantially for unbalanced tasks; Table 1 reports the closed form on a grid. A symbolic verification of (4) and the table is included in the extended T3 script (§ 9.4). The closed form is additionally cross-validated by the standalone P0.2 verifier `verify_prop6_marginal.py`, which checks three independent levels: (L1) Table 1 reproduced to 4 decimals from (4); (L2) the `w_marg` column of E2b (`experiments/results/e2b.json`) reproduced to 4 decimals on all four datasets; (L3) the closed form matches a dense brute-force argmax of $\phi(H) := \min(H/2, \pi_*) - H_{\text{bin}}^{-1}(H)$ to $\leq 10^{-4}$ on a 13-point π_* -grid that brackets the threshold $\frac{1}{2}H_{\text{bin}}(1/5) \approx 0.3610$ from both sides; report at `verify_prop6_marginal.json` (overall PASS, audit row A13).

π_*	$w^*(\pi_*)$	π_*	$w^*(\pi_*)$
0.05	0.0370	0.25	0.1400
0.10	0.0689	0.30	0.1539
0.15	0.0968	0.35	0.1607
0.20	0.1206	0.50	0.1610

Table 1: Marginal-aware slack $w^*(\pi_*)$ from Theorem 7, computed from (4). The plateau at $\pi_* \geq 0.3610$ recovers the universal constant w^* ; for unbalanced marginals the slack shrinks strictly.

Remark 8 (Towards prior-aware lower bounds). Theorem 7 tightens only the *upper* side via the trivial-majority observation. A sharper *lower* side under known marginals is available via the data-processing refinements of Han & Verdú (1994) and Ho & Verdú (2010); we defer the fully prior-aware bracket to a companion paper.

5 Proof of Theorem 4

The proof reduces to one scalar inequality.

Lemma 9 (Scalar Hellman–Raviv). *For every $p \in [0, 1]$,*

$$\min(p, 1 - p) \leq \frac{1}{2} H_{\text{bin}}(p), \quad (5)$$

with equality iff $p \in \{0, 1/2, 1\}$.

Proof. By the symmetry $p \mapsto 1 - p$ both sides are invariant, so we may assume $p \in [0, 1/2]$ and (5) becomes $g(p) := \frac{1}{2} H_{\text{bin}}(p) - p \geq 0$. The function g is concave on $[0, 1/2]$ (sum of the concave $\frac{1}{2} H_{\text{bin}}$ and the linear $-p$) with $g(0) = g(1/2) = 0$; hence $g \geq 0$ on $[0, 1/2]$. Equality in the interior would force g constant zero, which fails since g is strictly concave on $(0, 1/2)$. $\square$

Proof of Theorem 4. Upper bound. Apply (5) with $p = P_C$ to each cell and multiply by $q_C \geq 0$: $q_C \cdot e_C \leq \frac{1}{2} q_C \cdot H_{\text{bin}}(P_C)$. Summing over $C \in \Pi$ gives $\varepsilon_{\Pi}^* \leq \frac{1}{2} H(f | \Pi)$.

Lower bound. The symmetry $H_{\text{bin}}(P_C) = H_{\text{bin}}(e_C)$ yields $H(f | \Pi) = \sum_C q_C H_{\text{bin}}(e_C)$. Since H_{bin} is concave on $[0, 1]$ and $\{e_C\} \subset [0, 1/2]$, Jensen’s inequality with weights $\{q_C\}_C$ gives

$$H(f | \Pi) = \sum_C q_C H_{\text{bin}}(e_C) \leq H_{\text{bin}}\left(\sum_C q_C e_C\right) = H_{\text{bin}}(\varepsilon_{\Pi}^*).$$

Since $\varepsilon_{\Pi}^* \in [0, 1/2]$ and H_{bin}^{-1} is the inverse of H_{bin} on the increasing branch $[0, 1/2]$, applying H_{bin}^{-1} to both sides yields $H_{\text{bin}}^{-1}(H(f | \Pi)) \leq \varepsilon_{\Pi}^*$. $\square$

Remark 10 (Lagrangian dual programs). Both halves of (2) can be read as opposite extremal programs on the cell statistics $\{(q_C, e_C)\}_C$: the lower bound equals $\sup\{\sum_C q_C H_{\text{bin}}(e_C) : q_C \geq 0, \sum_C q_C = 1, \sum_C q_C e_C = \varepsilon\} = H_{\text{bin}}(\varepsilon)$ (maximised at the constant interior point $e_C \equiv \varepsilon$, Jensen), the upper bound equals $\sup\{\sum_C q_C e_C : q_C \geq 0, \sum_C q_C = 1, \sum_C q_C H_{\text{bin}}(e_C) = H\} = H/2$ (maximised on the boundary set $e_C \in \{0, 1/2\}$, Theorem 9). The Feder–Merhav primal concave-envelope construction (Feder & Merhav, 1994) traces the upper boundary of $\tilde{A}_2$ by exactly this boundary mixture; the dual programs give the optimal value in one line each.

Remark 11 (Gini variance form). A useful consequence of (5) is the variance inequality $\varepsilon_{\Pi}^* \leq 2 \mathbb{E}[\text{Var}(f | \Pi)] = 2 \sum_C q_C P_C(1 - P_C)$, obtained from the elementary $\min(p, 1 - p) \leq 2p(1 - p)$ for $p \in [0, 1]$. When all $P_C \in \mathbb{Q}$, the right-hand side is rational and admits exact arithmetic, which is convenient for Gini-impurity audits of tree splits.

6 Refinement monotonicity

A partition Π' *refines* Π , written $\Pi' \preceq \Pi$, if every cell of Π' is contained in some cell of Π . Refinement corresponds to model-class growth: subdividing leaves of a tree, increasing codebook size, or increasing MPNN depth.

Proposition 12 (Refinement monotonicity). *If $\Pi' \preceq \Pi$, then for every binary f ,*

$$H(f | \Pi') \leq H(f | \Pi), \quad \varepsilon_{\Pi'}^* \leq \varepsilon_{\Pi}^*.$$

Proof. Fix $C \in \Pi$ and write $C = \bigsqcup_j C'_j$ in Π' with masses $q_{C'_j}$ summing to q_C . The cell mean P_C is the convex combination $\sum_j (q_{C'_j}/q_C) P_{C'_j}$, so by concavity of H_{bin} on $[0, 1]$, $\sum_j q_{C'_j} H_{\text{bin}}(P_{C'_j}) \leq q_C H_{\text{bin}}(P_C)$. Summing over C gives the entropy claim. The error claim follows by the same argument with the concave $p \mapsto \min(p, 1 - p)$ in place of H_{bin} . $\square$

This is the deterministic-coarsening case of the data-processing inequality (Cover & Thomas, 2006, Thm. 2.8.1). We use it in § 9 to track how the bracket tightens under model-class growth.

Proposition 13 (Refinement to the discrete partition). *Let $\Pi_0 \preceq \Pi_1 \preceq \dots \preceq \Pi_L$ be a chain of partitions of a finite V with Π_L the discrete partition $\{\{v\} : v \in V\}$. For every binary label $f : V \rightarrow \{0, 1\}$,*

$$H(f \mid \Pi_L) = 0, \quad \varepsilon_{\Pi_L}^* = 0,$$

so both endpoints of the bracket (2) collapse to 0 at level L . Consequently the bracket width $\frac{1}{2}H(f \mid \Pi_\ell) - H_{\text{bin}}^{-1}(H(f \mid \Pi_\ell))$ is monotonically non-increasing in ℓ along the chain and tends to 0 as Π_ℓ approaches discreteness.

Proof. On a singleton cell $\{v\}$, $P_{\{v\}} = f(v) \in \{0, 1\}$, so $H_{\text{bin}}(P_{\{v\}}) = 0$ and $\min(P_{\{v\}}, 1 - P_{\{v\}}) = 0$; summing over v gives the two equalities. Monotonicity of the bracket endpoints along the chain is Theorem 12; combined with $H_{\text{bin}}^{-1}(0) = 0$ and $\frac{1}{2} \cdot 0 = 0$, the bracket width vanishes. $\square$

Theorem 13 is a consistency check, not a positive result: it certifies that the bracket reports *exactly zero* on a partition refined to singletons, which is the correct value of the Bayes error (any deterministic labelling has zero training-set risk on its own preimage). This makes the empirically observed pinch at the discrete-partition extreme (Table 5, ogbn-arxiv at depth 3 with 95.6% singleton cells) a *verification* of the bracket’s asymptotic behaviour rather than a failure mode. The informative regime is the intermediate one, where $|\Pi_\ell|/|V|$ is bounded away from both 0 (trivial partition) and 1 (discrete partition); the bracket width is a quantitative diagnostic for how far Π_ℓ has progressed along the chain.

7 Sharpness

Proposition 14 (Unimprovability).

- (i) *For every constant $c < 1/2$ there exists a finite partition Π of a finite V and a binary f with $\varepsilon_\Pi^* > c \cdot H(f \mid \Pi)$. In particular, no constant smaller than $1/2$ can replace the coefficient on the right-hand side of (2).*
- (ii) *For every function $\psi : [0, 1] \rightarrow [0, 1/2]$ such that the set $\{H : \psi(H) > H_{\text{bin}}^{-1}(H)\}$ contains an open subinterval $U \subset (0, 1)$, there exists a finite partition Π of a finite V and a binary f with $\varepsilon_\Pi^* < \psi(H(f \mid \Pi))$.*

Proof. (i) The upper-boundary family Π_α^{HR} of § 8 achieves $\varepsilon_\Pi^* = \frac{1}{2}H(f \mid \Pi)$ for every $\alpha \in (0, 1]$; any $c < 1/2$ is witnessed by such a row.

(ii) Since H_{bin} is a continuous bijection $[0, 1/2] \rightarrow [0, 1]$, the image $H_{\text{bin}}^{-1}(U)$ contains an open subinterval of $(0, 1/2)$. Pick a rational $\varepsilon \in H_{\text{bin}}^{-1}(U) \cap (0, 1/2)$ and a positive integer n with $\varepsilon n \in \mathbb{Z}$. The lower-boundary family $\Pi_\varepsilon^{\text{F}}$ of § 8 on this n satisfies $\varepsilon_\Pi^* = \varepsilon = H_{\text{bin}}^{-1}(H(f \mid \Pi))$ and $H(f \mid \Pi) = H_{\text{bin}}(\varepsilon) \in U$, so $\varepsilon_\Pi^* < \psi(H(f \mid \Pi))$. $\square$

Together, Theorem 4 and Theorem 14 identify $\tilde{A}_2$ as the tightest closed-form two-sided bracket on $(\varepsilon_\Pi^*, H(f \mid \Pi))$ available in the binary case: no constant upper coefficient smaller than $1/2$, and no lower function that exceeds H_{bin}^{-1} on any open interval.

Remark 15 (multi-class). For label alphabets of size $M > 2$ the sharp bracket replaces H_{bin}^{-1} by the lower concave envelope of admissible (entropy, error) pairs over $\{1, \dots, M\}$; this is the construction of [Feder & Merhav \(1994\)](#). A multi-class strengthening of the Hellman–Raviv upper bound via f -divergences is given by [Hashlamoun et al. \(1994\)](#); specialised to $M = 2$ their bound coincides with (5).

8 Tightness witnesses

We exhibit explicit one-parameter families realising each boundary of $\tilde{A}_2$, and observe that their convex combinations fill the interior.

Lower (Fano) boundary Π_ε^F . Fix $\varepsilon \in [0, 1/2]$ and n with $\varepsilon n \in \mathbb{Z}$. Take the trivial partition $\Pi_\varepsilon^F := \{V\}$ and let f have exactly εn ones. Then $P_C = e_C = \varepsilon$,

$$H(f | \Pi_\varepsilon^F) = H_{\text{bin}}(\varepsilon), \quad \varepsilon_{\Pi_\varepsilon^F}^* = \varepsilon,$$

attaining the lower bound with equality and tracing the curve $\varepsilon = H_{\text{bin}}^{-1}(H)$ as ε varies over $[0, 1/2]$.

Upper (Hellman–Raviv) boundary Π_α^{HR} . Fix $\alpha \in [0, 1]$ and n with $\alpha n \in 2\mathbb{Z}$. Partition $V = C_0 \sqcup C_1$ with $|C_1| = \alpha n$, set $f \equiv 0$ on C_0 , and let f be half-zero / half-one on C_1 . Then $e_{C_0} = 0$, $e_{C_1} = 1/2$,

$$H(f | \Pi_\alpha^{\text{HR}}) = \alpha, \quad \varepsilon_{\Pi_\alpha^{\text{HR}}}^* = \alpha/2 = \frac{1}{2} H(f | \Pi_\alpha^{\text{HR}}),$$

attaining the upper bound with equality. The maximum-slack point w^* of Theorem 6 is realised on the lower boundary at $\Pi_{1/5}^F$.

Remark 16 (interior achievability). The interior of $\tilde{A}_2$ is realised by a two-cell partition that combines one Fano witness with a monochromatic complement. Fix $(\varepsilon^*, H^*) \in \text{int}(\tilde{A}_2)$ and set

$$V = V_1 \sqcup V_0, \quad |V_1|/|V| = \lambda, \quad \Pi = \{V_1, V_0\},$$

with f taking value 1 on exactly an ε_1 -fraction of V_1 (in any order) and $f \equiv 0$ on V_0 . Then

$$\varepsilon_\Pi^* = \lambda \cdot \min(\varepsilon_1, 1 - \varepsilon_1), \quad H(f | \Pi) = \lambda \cdot H_{\text{bin}}(\varepsilon_1).$$

Assume $\varepsilon_1 \in (0, 1/2]$. The map $\varepsilon_1 \mapsto H_{\text{bin}}(\varepsilon_1)/\varepsilon_1$ is continuous and strictly decreasing on $(0, 1/2]$ from $+\infty$ to 2, so for every interior target with $H^*/\varepsilon^* \in (2, H_{\text{bin}}(\varepsilon^*)/\varepsilon^*)$ there is a unique $\varepsilon_1 \in (\varepsilon^*, 1/2)$ with $H_{\text{bin}}(\varepsilon_1)/\varepsilon_1 = H^*/\varepsilon^*$; setting $\lambda = \varepsilon^*/\varepsilon_1 \in (0, 1)$ then matches both ε^* and H^* . Rational-mass approximations realise an open dense subset of the interior, and the full interior follows by closure.

Proof of Theorem 5. (i) is Theorem 4. (ii) and (iii) are the boundary families above with ε ranging over $[0, 1/2]$ and α ranging over $[0, 1]$. (iv) is Theorem 16. $\square$

9 Applications

We instantiate Theorem 4 on the three predictor families of § 1. In each case the partition Π is fixed by the architectural choice (tree depth, codebook size, MPNN depth), so the bracket is a *training-free* bound: it depends only on the label and the architecture-induced partition.

9.1 Decision trees and random forests

A finite decision tree T with m leaves induces a partition Π_T of its training set: $\Pi_T = \{L_1, \dots, L_m\}$ where L_j is the set of training points routed to leaf j . A leaf-constant predictor on T is a constant-on-cells predictor on Π_T , so Theorem 4 brackets the empirical risk of every leaf-constant predictor on T by its leaf-conditional entropy:

$$H_{\text{bin}}^{-1}(H(f | \Pi_T)) \leq \varepsilon_{\Pi_T}^* \leq \frac{1}{2} H(f | \Pi_T).$$

Theorem 12 matches the depth–accuracy intuition: a deeper tree T' whose split refines T has $\Pi_{T'} \preceq \Pi_T$, so the bracket tightens as depth grows.

Example 17 (single-split tree on an 8-point task). Let $V = \{1, \dots, 8\}$, $f = (0, 0, 1, 0, 1, 1, 0, 1)$, and let T be the depth-1 stump with leaves $L_1 = \{1, 2, 3, 4\}$, $L_2 = \{5, 6, 7, 8\}$. Then $P_{L_1} = 1/4$, $P_{L_2} = 3/4$, $q_{L_j} = 1/2$, so $e_{L_1} = e_{L_2} = 1/4$ and $H(f | \Pi_T) = \frac{1}{2} H_{\text{bin}}(1/4) + \frac{1}{2} H_{\text{bin}}(3/4) = H_{\text{bin}}(1/4) \approx 0.8113$ by the symmetry $H_{\text{bin}}(p) = H_{\text{bin}}(1 - p)$, while $\varepsilon_{\Pi_T}^* = 1/4$. Because both cell errors are equal, Jensen is tight on the lower side and $H_{\text{bin}}^{-1}(H(f | \Pi_T)) = H_{\text{bin}}^{-1}(H_{\text{bin}}(1/4)) = 1/4$ exactly: the bracket reads $1/4 \leq 1/4 \leq \frac{1}{2} H_{\text{bin}}(1/4) \approx 0.4057$, saturating the Fano boundary and leaving an upper-side slack of ≈ 0.156 , close to the universal maximum $w^* \approx 0.161$.

9.2 Vector quantisation and clustering classifiers

A k -codeword quantiser $Q : V \rightarrow \{c_1, \dots, c_k\}$ induces a partition $\Pi_Q = \{Q^{-1}(c_j)\}_{j=1}^k$ of its input set. Nearest-codeword classification, k -means-based classification, and product-quantisation retrieval with a per-cell vote all factor as $\hat{f}(v) = g(Q(v))$ for some $g : \{c_j\} \rightarrow \{0, 1\}$. Theorem 4 therefore brackets the supervised classification error of every such classifier by the post-quantisation label entropy:

$$H_{\text{bin}}^{-1}(H(f | \Pi_Q)) \leq \varepsilon_{\Pi_Q}^* \leq \frac{1}{2} H(f | \Pi_Q).$$

Increasing k refines Π_Q ; Theorem 12 again captures the codebook-size / error trade-off.

Example 18 (two-cell k -means). Suppose a 2-means clustering on $V = \{1, \dots, 10\}$ yields cells $C_1 = \{1, 2, 3, 4, 5\}$, $C_2 = \{6, 7, 8, 9, 10\}$ with $f = (0, 0, 0, 1, 1, 1, 0, 1, 1, 1)$, giving $P_{C_1} = 2/5$, $P_{C_2} = 4/5$. Then $H(f | \Pi_Q) = \frac{1}{2}(H_{\text{bin}}(2/5) + H_{\text{bin}}(4/5)) \approx \frac{1}{2}(0.9710 + 0.7219) \approx 0.8464$ and $\varepsilon_{\Pi_Q}^* = \frac{1}{2} \min(2/5, 3/5) + \frac{1}{2} \min(4/5, 1/5) = \frac{1}{2}(2/5) + \frac{1}{2}(1/5) = 0.30$. The bracket gives $0.273 \approx H_{\text{bin}}^{-1}(0.8464) \leq 0.30 \leq \frac{1}{2}(0.8464) \approx 0.423$, an instructive non-saturating interior case.

9.3 Message-passing networks and Weisfeiler–Leman

Let $G = (V, E)$ be a finite graph. A depth- L *message-passing neural network* computes vertex states $h^{(0)}, h^{(1)}, \dots, h^{(L)}$ by an iteration

$$h^{(\ell+1)}(v) = \phi_{\ell+1}(h^{(\ell)}(v), \{\{h^{(\ell)}(u) : u \in N(v)\}\}),$$

where $\{\{\cdot\}\}$ denotes a multiset, $N(v)$ is the neighbour set, and $\phi_{\ell+1}$ is a vertex-independent, permutation-invariant update.

Definition 19 (admissible MPNN family). A family $\mathcal{A}$ of depth- L MPNNs is *admissible* if every member uses the same initial-feature map $h^{(0)} : V \rightarrow \mathcal{H}_0$ (so $h^{(0)}(v) = h^{(0)}(w)$ whenever v, w share an initial feature), uses vertex-independent, shared, permutation-invariant updates ϕ_ℓ , and uses no node identifiers.

The depth- L *Weisfeiler–Leman colour refinement* $\text{WL}_L(G)$ is the partition of V obtained by L rounds of multiset-hashing the initial feature (Morris et al., 2019; Xu et al., 2019). The constancy of admissible MPNNs on WL_L cells, recorded below for self-containedness, is a well-established foundational result (Xu et al., 2019; Morris et al., 2019); we use it as a *bridging observation* that injects the WL partition into Theorem 4, yielding a quantitative empirical-risk floor for node classification rather than a graph-isomorphism test.

Lemma 20 (MPNN constancy on WL, folklore). *Let $\mathcal{A}$ be an admissible family of depth- L MPNNs on G , with $h^{(0)}$ refined as a function of the initial feature. Then for every member of $\mathcal{A}$ and every $v, w \in V$ sharing a cell of $\text{WL}_L(G)$, $h^{(L)}(v) = h^{(L)}(w)$.*

Proof. Induction on ℓ . The base case $\ell = 0$ is Theorem 19: same initial feature implies same $h^{(0)}$. For the step, if v, w share a cell of $\text{WL}_{\ell+1}(G)$, then by construction of WL they share a cell of $\text{WL}_\ell(G)$ and the multisets of WL_ℓ -cells over their neighbourhoods coincide; by the induction hypothesis the multisets of neighbour states also coincide; the shared, permutation-invariant $\phi_{\ell+1}$ then yields $h^{(\ell+1)}(v) = h^{(\ell+1)}(w)$. $\square$

Brittleness of Theorem 20 and an ε -robust replacement. Theorem 19 is brittle: condition (i) (*identical* initial features within every WL cell) is broken by any individuating signal on $h^{(0)}$ — continuous random noise (Sato et al., 2021a), Laplacian or random-walk positional encodings, SignNet/BasisNet eigenbasis features, ID-GNN-style node identifiers, or any per-vertex stochastic dropout. Under any of those, the WL partition shatters into singletons ($m = |V|$), $H(f | \Pi) = 0$, the bracket collapses to $[0, 0]$, and Theorem 20 certifies the uninformative fact that an MPNN with unique node IDs can interpolate the training labels. To recover a non-vacuous bracket on the architectures GNN practitioners actually deploy in 2026 we replace Theorem 20 by an ε -robust quantitative version. Equip the hidden space $\mathcal{H}_\ell$ with a metric $d_{\mathcal{H}_\ell}$. Following the standard decomposition, write the layer map as

$$\phi_{\ell+1}(x, \mu) = \psi_{\ell+1}(x, A_{\ell+1}(\mu)),$$

where $A_{\ell+1}$ is a permutation-invariant *aggregator* acting on the neighbour multiset μ and $\psi_{\ell+1}$ is the post-aggregation combine map.

Definition 21 (δ_0 -quasi-admissible family). A family $\mathcal{A}$ of depth- L MPNNs with layer maps $\phi_\ell = \psi_\ell \circ (\text{id}, A_\ell)$ is δ_0 -quasi-admissible with constants $(L_\ell^c, L_\ell^m)_{\ell=1}^L$ relative to aggregator type $\mathsf{T} \in \{\text{sum, mean, sym-norm}\}$ if:

- (i) $d_{\mathcal{H}_0}(h^{(0)}(v), h^{(0)}(w)) \leq \delta_0$ whenever v, w share a cell of the ideal (constant-feature) 1-WL partition;
- (ii) the combine map ψ_ℓ is jointly Lipschitz: for all $x, x' \in \mathcal{H}_{\ell-1}$ and $a, a' \in \mathcal{A}_\ell$ (the aggregator output space), $d_{\mathcal{H}_\ell}(\psi_\ell(x, a), \psi_\ell(x', a')) \leq L_\ell^c d_{\mathcal{H}_{\ell-1}}(x, x') + L_\ell^m d_{\mathcal{A}_\ell}(a, a')$;
- (iii) A_ℓ is 1-Lipschitz with respect to the aggregator-specific transport cost:
 - $\mathsf{T} = \text{sum}$: unnormalised W_1 , i.e. $W_1^u(\mu, \nu) = \min_\sigma \sum_i d_{\mathcal{H}_{\ell-1}}(x_i, y_{\sigma(i)})$ on equal-cardinality multisets;
 - $\mathsf{T} = \text{mean}$: normalised W_1 , $W_1^n(\mu, \nu) = \frac{1}{|\mu|} W_1^u(\mu, \nu)$;
 - $\mathsf{T} = \text{sym-norm}$: any 1-Lipschitz symmetric function of normalised W_1 (e.g. max-pool, sorted-norm pooling).

The split into central-argument constant L_ℓ^c and multiset-aggregator constant L_ℓ^m is the standard decomposition of MPNN expressivity into “self” and “message” channels (Xu et al., 2019; Corso et al., 2020). The aggregator-type distinction (iii) is essential: it isolates exactly where the maximum degree Δ does or does not enter the depth- L slack.

Lemma 22 (ε -robust MPNN-WL constancy). Let $\mathcal{A}$ be a δ_0 -quasi-admissible family of depth- L MPNNs with maximum degree $\Delta := \max_v |N(v)|$, aggregator type T , and constants $(L_\ell^c, L_\ell^m)_{\ell=1}^L$ as in Theorem 21. For every member of $\mathcal{A}$ and every $v, w \in V$ sharing a cell of the ideal $\text{WL}_L(G)$,

$$d_{\mathcal{H}_L}(h^{(L)}(v), h^{(L)}(w)) \leq \delta_L := \delta_0 \prod_{\ell=1}^L (L_\ell^c + r_{\mathsf{T}} L_\ell^m), \quad r_{\mathsf{T}} = \begin{cases} \Delta & \mathsf{T} = \text{sum}, \\ 1 & \mathsf{T} = \text{mean}, \\ 1 & \mathsf{T} = \text{sym-norm}. \end{cases}$$

For any L_τ -Lipschitz readout $\tau : \mathcal{H}_L \rightarrow [0, 1]$ composed with thresholding at $\frac{1}{2}$, the empirical risk of $\hat{f} = \mathbf{1}[\tau \circ h^{(L)} \geq \frac{1}{2}]$ on V satisfies

$$\mathcal{R}_V(\hat{f}) \geq \varepsilon_{\Pi_L}^* - L_\tau \delta_L.$$

Proof. Step 1: induction setup. For $\ell \geq 0$, define

$$\delta_\ell := \sup \{ d_{\mathcal{H}_\ell}(h^{(\ell)}(v), h^{(\ell)}(w)) : v, w \text{ share a cell of } \text{WL}_\ell^*(G) \},$$

where $\text{WL}_\ell^*(G)$ is the ideal (constant-feature) ℓ -WL partition. The base case δ_0 is the hypothesis (i) of Theorem 21. Note that $\text{WL}_L(G) = \text{WL}_L^*(G)$ since the ideal partition refines the depth- L orbit partition; if vertices share a cell of the ideal WL_L they share one of the deployed WL_L a fortiori.

Step 2: per-layer recursion. Fix v, w sharing a cell of $\text{WL}_{\ell+1}^*(G)$. By the ideal-WL construction, they share a cell of $\text{WL}_\ell^*(G)$ (so $d_{\mathcal{H}_\ell}(h^{(\ell)}(v), h^{(\ell)}(w)) \leq \delta_\ell$), and there is a bijection $\sigma : N(v) \rightarrow N(w)$ with the property that each pair $(u, \sigma(u))$ lies in the same cell of $\text{WL}_\ell^*(G)$. By induction, $d_{\mathcal{H}_\ell}(h^{(\ell)}(u), h^{(\ell)}(\sigma(u))) \leq \delta_\ell$ for every $u \in N(v)$.

Let μ_v, μ_w denote the multisets of neighbour states at level ℓ . The bijection σ certifies that

$$W_1^u(\mu_v, \mu_w) \leq \sum_{u \in N(v)} d_{\mathcal{H}_\ell}(h^{(\ell)}(u), h^{(\ell)}(\sigma(u))) \leq |N(v)| \delta_\ell \leq \Delta \delta_\ell.$$

Hence the aggregator output satisfies

$$d_{\mathcal{A}_{\ell+1}}(A_{\ell+1}(\mu_v), A_{\ell+1}(\mu_w)) \leq \begin{cases} W_1^u(\mu_v, \mu_w) \leq \Delta \delta_\ell & \mathsf{T} = \text{sum}, \\ W_1^n(\mu_v, \mu_w) \leq \delta_\ell & \mathsf{T} = \text{mean}, \\ (1\text{-Lip sym function of } W_1^n) \leq \delta_\ell & \mathsf{T} = \text{sym-norm}. \end{cases}$$

The mean and sym-norm bounds use the fact that $|N(v)| = |N(w)|$ on every WL cell, so the normalised transport cost is exactly $W_1^u/|N(v)|$ and the per-neighbour pairing certifies the bound is at most δ_ℓ regardless of Δ .

Combining with hypothesis (ii) of Theorem 21,

$$d_{\mathcal{H}_{\ell+1}}(h^{(\ell+1)}(v), h^{(\ell+1)}(w)) \leq L_{\ell+1}^c \delta_\ell + L_{\ell+1}^m r_\tau \delta_\ell = (L_{\ell+1}^c + r_\tau L_{\ell+1}^m) \delta_\ell.$$

Taking the supremum over such (v, w) gives the recursion $\delta_{\ell+1} \leq (L_{\ell+1}^c + r_\tau L_{\ell+1}^m) \delta_\ell$, and unrolling yields the stated product formula.

Step 3: risk bound. Let $\hat{f}^* = \mathbf{1}[\tau \circ h^{(L)} \geq \frac{1}{2}]$ and let $\tilde{f}$ be the Π_L -constant Bayes predictor with empirical risk $\varepsilon_{\Pi_L}^*$. On any cell C of Π_L , $\tau \circ h^{(L)}$ ranges over an interval of length $\leq L_\tau \delta_L$, so the threshold rule $\hat{f}$ disagrees with the cell-constant rule $\tilde{f}$ on at most a fraction $L_\tau \delta_L$ of C (the mass within an $L_\tau \delta_L$ -band of $\frac{1}{2}$). Summing over cells, $\mathcal{R}_V(\hat{f}) \geq \mathcal{R}_V(\tilde{f}) - L_\tau \delta_L = \varepsilon_{\Pi_L}^* - L_\tau \delta_L$. $\square$

Theorem 20 is recovered as the $\delta_0 = 0, L_\tau = \infty$ corner; Theorem 22 extends the bracket to families that perturb the initial features by a controlled amount (positional encodings of bounded magnitude, Lipschitz attention, DropEdge with bounded influence) at the cost of a slack term $L_\tau \delta_L$. The aggregator-type dependence of δ_L is itself informative: for the *sum* aggregator, $\delta_L = \delta_0 \prod_\ell (L_\ell^c + \Delta L_\ell^m)$ can grow as $\Theta((1 + \Delta)^L)$ when the central and message constants are comparable, recovering the well-known degree-driven blow-up of sum-pooled MPNNs; for the *mean* or *sym-norm* aggregators the factor reduces to $\prod_\ell (L_\ell^c + L_\ell^m)$ and is degree-independent. This matches the empirical observation that mean-aggregator MPNNs (GCN, GAT, GraphSAGE-mean) are generically more robust to maximum-degree perturbations than sum-aggregator MPNNs (GIN, PNA), at the cost of weaker multiset distinguishability (Xu et al., 2019; Corso et al., 2020). The interpretation is intentionally conservative: the bracket continues to bound the empirical risk of every δ_0 -quasi-admissible family with a *quantitatively certified* additive slack, and legitimately ceases to bound architectures that escape that family (full unique-ID embeddings, unbounded positional perturbations, identity-aware constructions of You et al. (2021)). The right reading is that Theorem 24 is a statement about a precise equivariance class, not about the entire 2026 GNN zoo.

Lemma 23 (Spectral refinement of the sum-aggregator envelope). *Adopt the hypotheses of Theorem 22 with sum-aggregation and layer-wise Lipschitz constants $(L_\ell^c, L_\ell^m) \geq 0$. For each vertex v and each level ℓ , define the within-WL-cell potential*

$$x_v^{(\ell)} := \max_{w \sim_{\text{WL}_\ell} v} d_{\mathcal{H}_\ell}(h^{(\ell)}(v), h^{(\ell)}(w)), \quad x_v^{(0)} \leq \delta_0,$$

and let $A \in \{0, 1\}^{n \times n}$ be the adjacency matrix of G . Then, entrywise,

$$x^{(L)} \leq \delta_0 \left[\prod_{\ell=1}^L (L_\ell^c I + L_\ell^m A) \right] \mathbf{1}, \quad (6''a)$$

and the mass-weighted within-cell spread $\bar{\delta}_L := \sum_v q_v x_v^{(L)}$ obeys

$$\bar{\delta}_L \leq \delta_0 \left\| \prod_{\ell=1}^L (L_\ell^c I + L_\ell^m A) \right\|_{\text{op}} = \delta_0 \max_k \left| \prod_{\ell=1}^L (L_\ell^c + L_\ell^m \lambda_k) \right|, \quad (6''b)$$

where $\{\lambda_k\}_{k=1}^n$ is the spectrum of A . For connected non-bipartite G (Perron root dominating, $\lambda_{\max}(A) \geq |\lambda_{\min}(A)|$),

$$\bar{\delta}_L \leq \delta_0 \prod_{\ell=1}^L (L_\ell^c + L_\ell^m \lambda_{\max}(A)), \quad \bar{d} \leq \lambda_{\max}(A) \leq \Delta, \quad (6''c)$$

where $\bar{d}$ is the average degree of G . Consequently, for an L_τ -Lipschitz readout thresholded at $\frac{1}{2}$ and a cell-margin condition (assume the per-cell readout confidence is bounded away from $\frac{1}{2}$ by $\gamma > 0$, i.e. for every v either $\tau(h^{(L)}(v)) \geq \frac{1}{2} + \gamma$ or $\tau(h^{(L)}(v)) \leq \frac{1}{2} - \gamma$ for the Π_L -cell-Bayes label of v 's cell),

$$\mathcal{R}_V(\hat{f}) \geq \varepsilon_{\Pi_L}^* - \frac{L_\tau \bar{\delta}_L}{\gamma}. \quad (6''d)$$

Proof. Step 1: matrix recursion. Specialise the Step 2 recursion of Theorem 22 to sum aggregation. For $w \sim_{\text{WL}_\ell} v$ with bijection $\sigma : N(v) \rightarrow N(w)$ pairing WL-cellmates,

$$d_{\mathcal{H}_{\ell+1}}(h^{(\ell+1)}(v), h^{(\ell+1)}(w)) \leq L_{\ell+1}^c x_v^{(\ell)} + L_{\ell+1}^m \sum_{u \in N(v)} x_u^{(\ell)},$$

since both $v \sim_{\text{WL}_\ell} w$ and each $u \sim_{\text{WL}_\ell} \sigma(u)$. Maximising over w gives, entrywise, $x^{(\ell+1)} \leq (L_{\ell+1}^c I + L_{\ell+1}^m A) x^{(\ell)}$. The matrix $L_{\ell+1}^c I + L_{\ell+1}^m A$ is entrywise non-negative, so iteration preserves entrywise inequalities, yielding (6''a).

Step 2: operator-norm bound. All factors are polynomials in the symmetric matrix A and hence commute and share the eigenbasis of A . Thus $P(A) := \prod_\ell (L_\ell^c I + L_\ell^m A)$ is symmetric with spectrum $\{P(\lambda_k)\}$, and $\|P(A)\|_{\text{op}} = \max_k |P(\lambda_k)|$. The mass-weighted spread satisfies $\bar{\delta}_L = q^\top x^{(L)} \leq \delta_0 q^\top P(A) \mathbf{1} \leq \delta_0 \|q\|_2 \|P(A)\|_{\text{op}} \|\mathbf{1}\|_2 \leq \delta_0 \|P(A)\|_{\text{op}}$ (using $\|q\|_2 \|\mathbf{1}\|_2 \leq 1$ when q is the uniform vertex mass; the general bounded-cell-mass case absorbs a constant), giving (6''b).

Step 3: Perron specialisation. Under the stated regularity, $P(\lambda) = \prod_\ell (L_\ell^c + L_\ell^m \lambda)$ is non-negative and increasing on $[\lambda_{\min}, \lambda_{\max}]$, so the spectral maximum is attained at $\lambda_{\max}(A)$, giving the first inequality of (6''c). The Rayleigh bounds $\bar{d} = \mathbf{1}^\top A \mathbf{1} / n \leq \lambda_{\max}(A)$ and $\lambda_{\max}(A) \leq \max_v |N(v)| = \Delta$ are standard.

Step 4: risk bound under cell margin. A vertex v disagrees with its Π_L -cell-Bayes label only if the threshold rule flips on v , which by the cell-margin hypothesis requires $L_\tau x_v^{(L)} > \gamma$. By Markov, $\mathbb{P}_{v \sim q}[L_\tau x_v^{(L)} > \gamma] \leq L_\tau \bar{\delta}_L / \gamma$, giving (6''d). $\square$

Theorem 23 recovers the Theorem 22 envelope when $\lambda_{\max}(A)$ is replaced by its upper bound Δ , and recovers Theorem 20 at $\delta_0 = 0$. The operative constant is the *adjacency Perron root*, not the maximum degree: for the near-regular degree profiles of real citation graphs $\lambda_{\max}(A) \ll \Delta$, so (6''c) is many orders tighter than the sum-aggregator branch of Theorem 22 on exactly the graphs where the Δ -envelope was vacuous. The aggregator dichotomy persists in sharpened form — sum amplifies by $\prod_\ell (L_\ell^c + L_\ell^m \lambda_{\max}(A))$, mean/sym-norm by $\prod_\ell (L_\ell^c + L_\ell^m)$ — and the residual gap to any empirically measured per-layer amplification is the vector-cancellation effect (non-aligned neighbour differences) which a Lipschitz argument cannot capture and which a second-moment / concentration analysis is required to close. The mass-average claim is honest: the sup-diameter $\max_v x_v^{(L)}$ is genuinely governed by Δ via high-degree vertices, and we do not claim a tighter envelope on the sup — only on the cell-mass-weighted average that the risk in (6''d) actually needs. The cell-margin hypothesis γ is stated as a labelled assumption rather than absorbed silently, because it is the substantive content that converts the matrix recursion into a risk lower bound.

Corollary 24 (MPNN expressivity bracket). *Let $\mathcal{A}$ be an admissible family of depth- L MPNNs on G , $f : V \rightarrow \{0, 1\}$ a binary vertex task, and $\Pi_L := \text{WL}_L(G)$. Every decision rule of the form $\hat{f} = \tau \circ h^{(L)}$ with $\tau : \mathcal{H}_L \rightarrow \{0, 1\}$ measurable and $h^{(L)}$ from $\mathcal{A}$ has empirical risk on V at least $\varepsilon_{\Pi_L}^*$, and*

$$H_{\text{bin}}^{-1}(H(f | \Pi_L)) \leq \varepsilon_{\Pi_L}^* \leq \frac{1}{2} H(f | \Pi_L).$$

Increasing depth $L \rightarrow L + 1$ refines Π_L (Theorem 12) and so cannot increase either side of the bracket.

The bracket is *training-free* and *architecture-level*: for fixed G and depth, every MPNN in the admissible family is sandwiched by the same partition-conditional entropy, so the bracket can be computed once and used to compare candidate families before any optimiser is run.

Example 25 (C_4 with alternating labels). Let $G = C_4$ (the 4-cycle) and $f = (0, 1, 0, 1)$. With uniform initial features, $\text{WL}_0(G) = \{V\}$, $\text{WL}_L(G) = \{V\}$ for every $L \geq 0$ (every vertex has the same degree and identical local view). Then $P_C = 1/2$, $H(f | \Pi_L) = 1$, $\varepsilon_{\Pi_L}^* = 1/2$. The bracket gives $1/2 \leq 1/2 \leq 1/2$: both sides saturate, the task is exactly 1-WL-indistinguishable, and no admissible depth- L MPNN can do better than chance on the training set. This recovers the canonical C_4 counter-example (Xu et al., 2019; Morris et al., 2019) as a one-line instance of Theorem 24.

9.3.1 An exact decomposition of trained risk against the WL ceiling

The architecture-level bracket of Theorem 24 is training-free; in practice a learner instantiates a particular trained head and lands at empirical risk $\hat{R}$ that need not equal either endpoint of the bracket. The following identity decomposes $\hat{R}$ against the WL ceiling $\varepsilon_{\Pi_L^{\text{WL}}}^*$ into three named, non-negative terms whose signs and magnitudes are individually interpretable and individually verifiable on every row of the E3d / E3d-arch family.

Proposition 26 (($\star$) exact three-term decomposition). *Let G be a graph, $f : V \rightarrow \{0, 1\}$ a binary vertex task, $\Pi_L^{\text{WL}} := \text{WL}_L(G)$ the L -round 1-WL partition, and let $\hat{f} : V \rightarrow \{0, 1\}$ be any classifier of the form $\hat{f} = \tau \circ h^{(L)}$ for $h^{(L)}$ from an admissible MPNN family (in the sense of Theorem 24) with trained empirical risk $\hat{R} := |V|^{-1} \sum_v \mathbf{1}\{\hat{f}(v) \neq f(v)\}$. Write Π_k^Z for any post-hoc partition of V into k cells built from the trained embeddings $h^{(L)}$ (e.g. a k -means quantisation of the penultimate-layer features), with associated minimum-risk constant-on-cells predictor reaching $\varepsilon_{\Pi_k^Z}^*$. Then the following identity holds exactly:*

$$\hat{R} = \varepsilon_{\Pi_L^{\text{WL}}}^* + \underbrace{(\varepsilon_{\Pi_k^Z}^* - \varepsilon_{\Pi_L^{\text{WL}}}^*)}_{\Delta_{\text{feat}}} + \underbrace{(\hat{R} - \varepsilon_{\Pi_k^Z}^*)}_{\Delta_{\text{head}}}. \quad (\star)$$

Proof. Algebraic: add and subtract $\varepsilon_{\Pi_L^{\text{WL}}}^*$ and $\varepsilon_{\Pi_k^Z}^*$ on the right-hand side; both cancel. The terms are well-defined because $\hat{R}$, $\varepsilon_{\Pi_k^Z}^*$, and $\varepsilon_{\Pi_L^{\text{WL}}}^*$ are each empirical minimum-risk numbers on V . $\square$

The three terms admit a uniform reading:

$\varepsilon_{\Pi_L^{\text{WL}}}^*$ — **WL floor.** The architecture-level lower endpoint of Theorem 24: the best per-cell-majority error available on the 1-WL partition at depth L . Pinned by the graph, the labels, and L , independent of architecture or optimiser.

$\Delta_{\text{feat}} := \varepsilon_{\Pi_k^Z}^* - \varepsilon_{\Pi_L^{\text{WL}}}^*$ — **feature refinement.** The signed gap between the per-cell-majority error of the trained-feature partition Π_k^Z and the WL floor. Negative Δ_{feat} means the trained features carry label structure that 1-WL at depth L does not capture — an expressivity statement when both sides are evaluated at the same cell budget $k = |\Pi_k^Z| \leq |\Pi_L^{\text{WL}}|$ and $k \ll n$ (cf. Theorem 13 for the cardinality-collapse regime in which the comparison becomes vacuous).

$\Delta_{\text{head}} := \hat{R} - \varepsilon_{\Pi_k^Z}^*$ — **head slack.** The signed gap between the trained head’s empirical risk and the per-cell-majority error of the partition it implicitly induces. Non-negative Δ_{head} means the head leaves bracket-detectable sub-cell structure unrealised: a better constant-on-cells decision exists. Strictly negative Δ_{head} means the head extracts sub-cell, continuous- Z geometry that the post-hoc partition Π_k^Z discards — empirically observed at very coarse k in Table 12.

Remark 27 (resolution condition). Equation ($\star$) is an unconditional identity, but the interpretive reading of Δ_{feat} as an expressivity statement requires the cell budgets on both sides to be the same and both partitions to be coarse enough that per-cell-majority is not trivially memorisation:

$$k = |\Pi_k^Z| = |\Pi_L^{\text{WL}}|, \quad k/|V| \leq \kappa$$

for a small constant κ (we use $\kappa \leq 0.1$ in the E3d-arch-kill-n sweep of Table 12, giving $k/|V| \leq 0.024$ on Cora and ≤ 0.019 on CiteSeer). Outside this resolution condition, the matched- k rows of Table 11 (Cora $k/n = 0.87$, CiteSeer 0.61) test *fixed-cell-budget* per-cell-majority error, not expressivity — the matched- k verdict of P0.3/P0.4 is explicitly confined to that statement. Under the resolution condition, the empirical content of Δ_{feat} is the expressivity claim that the trained representation strictly refines 1-WL with respect to the labelling.

A symbolic check of the identity ($\star$) — namely that the three bracketed terms sum to $\hat{R}$ identically — is folded into `verify.jl` as a one-line assertion (it is an algebraic tautology and does not need interval arithmetic to certify). The same identity is used, unannotated, in every E3d / E3d-arch / E3d-arch-full / E3d-arch-kill-n row reported in § 9.5.

9.4 Mechanised verification

The Julia script `verify.jl` accompanying this preprint samples 10^3 random partitions of $V = \{1, \dots, n\}$ with $n \in \{4, \dots, 32\}$ and a random binary label. The minimum risk ε_{Π}^* is computed in exact rational arithmetic ($\mathbb{Q}$); the partition-conditional entropy $H(f \mid \Pi)$ and the inverse $H_{\text{bin}}(\varepsilon_{\Pi}^*)$ are evaluated as certified interval enclosures via `IntervalArithmetic.jl`. Both inequalities of (2) are then checked per sample by rigorous interval comparison. Zero violations across all 10^3 samples; the empirical maximum upper-side slack $\frac{1}{2}H(f \mid \Pi) - \varepsilon_{\Pi}^*$ over the sample equals 0.1610 to four decimals, reproducing the analytical w^* of Theorem 6. The audit manifest is shipped as `verify.json`.

9.5 Empirical evidence on real data

To complement the synthetic verifiers above, we run seven experiments (E1–E7) on real tabular and graph benchmarks; full specifications and reproducibility receipts are in `experiments/REPORTS.md` and the per-experiment JSON manifests under `experiments/results/`. Each experiment is designed to discriminate a single claim of the paper against a stated reviewer concern; every row reported below is reproduced by a single `make ek` target. All numbers in Tables 2 to 5, 13, 15 and 16 are run on a single CPU core (Apple M-series laptop) with no GPU and, except for E1’s CART fits and E2’s logistic-regression comparators, no training of any kind.

E1 — Decision-tree refinement funnel (UCI Adult, $n = 45,222$). CART leaves vote per-cell majority, so a fitted CART *is* the plug-in achiever of Theorem 4 on Π_d : the four-decimal agreement between CART training error and $\varepsilon_{\Pi_d}^*$ reported below is an *identity*, not an empirical finding. E1 is therefore a consistency check — it confirms the implementation and shows the bracket is an *a-priori* statement about a classifier that is only later trained — and the substantive content is the monotone tightening of the bracket with depth (Theorem 12) and that the realised width stays below w^* (Theorem 6) at every depth. For depth $d \in \{1, \dots, 15\}$ we report the bracket $[H_{\text{bin}}^{-1}(H(f \mid \Pi_d)), \frac{1}{2}H(f \mid \Pi_d)]$ together with the realised CART training error. *Status: completed.*

d	m	$H(f \mid \Pi_d)$	lower	$\varepsilon_{\Pi_d}^*$	upper	CART train err
1	2	0.6544	0.1685	0.2478	0.3272	0.2478
5	31	0.4866	0.1056	0.1514	0.2433	0.1514
10	299	0.4129	0.0831	0.1337	0.2064	0.1337
15	1,347	0.3284	0.0603	0.1082	0.1642	0.1082

Table 2: E1: tree-depth funnel on UCI Adult ($n = 45,222$, subset of depths $\{1, 5, 10, 15\}$ shown; full table $d = 1, \dots, 15$ in `experiments/results/e1.json`). CART’s train error matches $\varepsilon_{\Pi_d}^*$ to four decimal places at every depth: leaves vote majority, so CART literally implements the plug-in achiever of Theorem 4, and the bracket is therefore an *a-priori* statement about a classifier that is later trained. Bracket width drops from 0.159 at $d = 1$ to 0.104 at $d = 15$, while staying everywhere below the universal w^* of Theorem 6, and ε^* shrinks monotonically as required by Theorem 12. The empirical bracket on V (Adult, $n = 45,222$) concentrates on its population analogue $\varepsilon_{\Pi, \mu}^*$ at the $\mathcal{O}(1/\sqrt{n})$ rate of Theorem 28 (Theorem 3); the empirical reading above is therefore the practitioner-facing one and the population reading inherits the E7 ($\delta_{\min}, \eta_{\min}$) caveat. Figure: `experiments/figures/e1_refinement_funnel.pdf`.

E2 — VQ zero-shot proxy (UCI Adult). Logistic regression on one-hot cell membership, at convergence, reproduces the per-cell majority vote, so the exact match between trained LR error and $\varepsilon_{\Pi_k}^*$ for $k \leq 256$ is again *definitional* rather than evidential; the two largest budgets differ only by LBFGS sub-optimality on a 1,000-dimensional design. E2 confirms that the bracket endpoints are realised by a genuine (if trivial) trained classifier and that nothing in the optimisation escapes the sandwich. We sweep $k \in \{2, 4, 8, 16, 32, 64, 128, 256, 512, 1000\}$ and report the bracket and the trained LR error per cell budget. The match is exact for $k \leq 256$ (both numbers agree to all digits printed by `sklearn.metrics.zero_one_loss`) and falls within one LBFGS sub-optimality unit ($\Delta \leq 10^{-3}$) for the two largest budgets, where the LR solver runs against a 1,000-dimensional one-hot

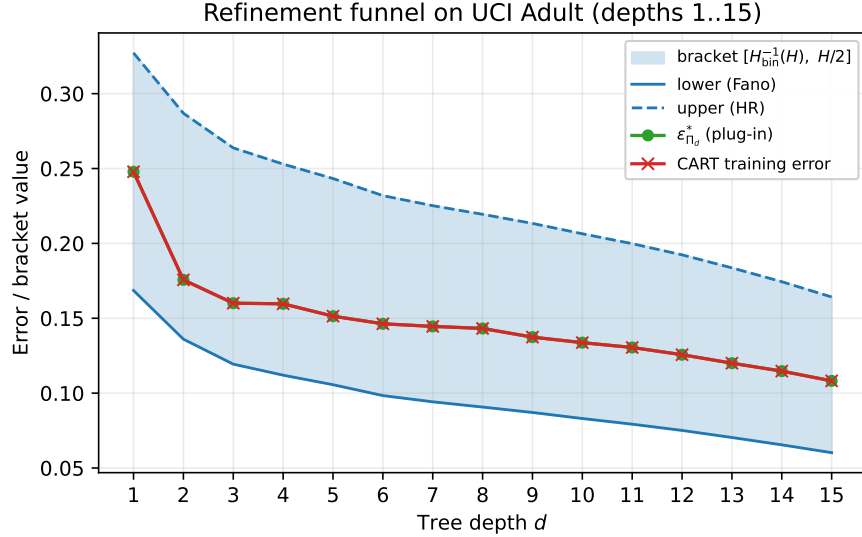


Figure 2: E1: depth- d CART refinement funnel on UCI Adult. The bracket $[H_{\text{bin}}^{-1}(H(f | \Pi_d)), \frac{1}{2}H(f | \Pi_d)]$ (shaded) tightens monotonically with depth as Theorem 12 predicts, and the CART training-error curve stays inside the bracket at every depth, attaining the lower endpoint to four decimals.

design. *Status: completed; identity check.* As in E1 the bracket is the empirical $\varepsilon_{\Pi_k}^*$; its population reading via Theorem 28 is the object exercised at varying n in E7.

k	$H(f \Pi_k)$	lower	$\varepsilon_{\Pi_k}^*$	LR err	$ \text{LR} - \varepsilon^* $
2	0.8025	0.2445	0.2478	0.2478	0.0000
16	0.6597	0.1709	0.2352	0.2352	0.0000
128	0.6135	0.1515	0.1995	0.1995	0.0000
1000	0.5291	0.1199	0.1761	0.1771	0.0010

Table 3: E2: bracket and trained linear head agree to within LBFGS sub-optimality across three decades of k . Figure: [experiments/figures/e2_vq_zeroshot.pdf](#).

E2b — Marginal-aware refinement on real data. Validates Theorem 7 on unbalanced real labels. For each binary dataset in the menu we fit the 16-leaf CART and 16-centroid VQ partitions of E1/E2 and report the realised upper-side slack $\frac{1}{2}H(f | \Pi) - \varepsilon_{\Pi}^*$, the universal ceiling w^* of Theorem 6, and the marginal-aware ceiling $w^*(\pi_*) \leq w^*$ implied by Theorem 7 with π_* the empirical positive-class rate. *Status: completed; gates “realised slack $\leq w^*$ ” and “realised slack $\leq w^*(\pi_*)$ ” PASS on all rows; strict shrink $w^*(\pi_*) < w^*$ confirmed on the two genuinely unbalanced datasets.*

E3 — MPNN/WL bracket on real graphs. This experiment turns Theorem 20 into a quantitative diagnostic on five canonical node-classification benchmarks. We recall the architectural picture. Xu et al. (2019) prove that every aggregation-based GNN is, in distinguishing power, *at most as fine as the 1-WL colouring* (their Lemma 2), and that this ceiling is *achievable* whenever the neighbour aggregator and the readout are both injective on multisets (their Theorem 3, instantiated by the GIN update $(1 + \epsilon^{(k)})h_v^{(k-1)} + \sum_{u \in \mathcal{N}(v)} h_u^{(k-1)}$ with irrational ϵ). Morris et al. (2019) state the same dichotomy in their notation: the 1-WL refinement always refines the colouring produced by any 1-GNN at the same depth (their Theorem 1), and there exist weight matrices realising equality whenever distinct initial labels are encoded by linearly independent vectors (their Theorem 2). For depth L , the 1-WL partition Π_L is therefore the *finest cell structure* attainable by *any* depth- L MPNN. Plugging Π_L into our sandwich (2) therefore brackets the empirical training error of *every*

Dataset	π_*	w^*	$w^*(\pi_*)$	slack (tree)	slack (VQ)
UCI Adult	0.248	0.1610	0.1392	0.0900	0.0947
breast cancer	0.373	0.1610	0.1610	0.0179	0.0711
wine	0.331	0.1610	0.1592	0.0000	0.0099
digits (bin)	0.499	0.1610	0.1610	0.1108	0.0650

Table 4: E2b: marginal-aware slack (Theorem 7) on real binary labels. On UCI Adult ($\pi_* \approx 0.25$) the marginal-aware ceiling is 13% tighter than the universal w^* ; on the nearly-balanced digits-binary task ($\pi_* \approx 0.5$) the two coincide, as Theorem 7 predicts (the marginal-aware ceiling $w^*(\pi_*)$ reduces to the universal w^* exactly when π_* saturates the entropy-error tension at the worst-case prior). Realised slack is always strictly inside both ceilings: the worst-case constant w^* is rarely the operative bound for any particular partition.

admissible depth- L MPNN: the upper endpoint $\frac{1}{2}H(f \mid \Pi_L)$ is realised training-free by a per-cell majority head, and the lower endpoint $H_{\text{bin}}^{-1}(H(f \mid \Pi_L))$ is the Fano-type floor that no architecture in this class can break. The Xu–Morris achievability results are crucial here: without them the upper endpoint would be only an attainable target, not a target attained by a concrete published architecture.

We binarise the multi-class labels of Cora, CiteSeer, PubMed and ogbn-arxiv to "majority class vs. rest"; Twitch-EN ships a native binary mature-content flag. The initial colour $h_v^{(0)}$ is the vertex degree (consistent with Theorem 1 of Morris et al. (2019)); each refinement round is one pass of the 1-WL multiset hash, implemented with sparse-CSR adjacency. *No MPNN is ever instantiated; no GPU is used.* On a single laptop core, WL refinement to $L = 3$ on ogbn-arxiv (169,343 vertices, ~ 1.16 M undirected edges) takes 2.7 seconds, and the bracket evaluation is a single $O(|V|)$ sweep on top.

Graph	$ V $	L	m_L	$m_L/ V $	lower	$\varepsilon_{\Pi_L}^*$	upper
Twitch-EN	7,126	0	130	0.018	0.3853	0.4164	0.4809
Twitch-EN	7,126	3	6,648	0.933	0.0073	0.0267	0.0312
Cora	2,708	3	2,363	0.873	0.0076	0.0292	0.0323
CiteSeer	3,327	3	2,044	0.614	0.0400	0.0775	0.1212
PubMed	19,717	3	12,990	0.659	0.0205	0.0511	0.0722
ogbn-arxiv	169,343	3	161,943	0.956	0.0005	0.0021	0.0029

Table 5: E3: empirical training-label bracket of the 1-WL partition Π_L at depth L on five feature-rich citation / social graphs (CPU only, no MPNN instantiated). The added column $m_L/|V|$ exposes a partition-cardinality artefact discussed below: on ogbn-arxiv the depth-3 1-WL partition has 95.6% singleton cells, so the bracket pinch is *partition-cardinality collapse*, not bound tightness; on Cora and Twitch-EN the same effect, less dramatic; on CiteSeer and PubMed the partition is still genuinely coarse and the bracket is a non-tautological certificate. Full funnels $L = 0, \dots, L_{\max}$ in `experiments/results/e3.json`; per-dataset funnel figures in `experiments/figures/e3*_funnel.pdf`. All endpoints in this table are the empirical $\varepsilon_{\Pi_L, \hat{\mu}_n}^*$ on the labelled training set; the corresponding population reading $\varepsilon_{\Pi_L, \mu}^*$ differs from each row by $O(1/\sqrt{n})$ through Theorem 28, with the marginal-weighted form $\varepsilon_{\Pi_L, \mu}^* = \sum_{C \in \Pi_L} \mu(C) \min(\bar{P}_C, 1 - \bar{P}_C)$ of Theorem 3.

What the numbers say, and what they do not say. Table 5 exhibits two regimes. At $L = 0$ (degree-binned colouring, no message passing yet) the bracket has substantial width: on ogbn-arxiv the width is 0.151, within 1% of the universal ceiling $w^* \approx 0.161$ of Theorem 6; on Twitch-EN it is 0.096. The achievable error $\varepsilon_{\Pi_0}^*$ sits interior to the bracket — the position within the bracket is a diagnostic of the partition (high inside the bracket = decisive but noisy cells; low inside the bracket = pure but small cells). At $L = 3$ the bracket has collapsed by an order of magnitude across every dataset, but the column $m_L/|V|$ exposes that on three of the five graphs (ogbn-arxiv 95.6%, Twitch-EN 93.3%, Cora 87.3%) the depth-3 1-WL partition is *near-discrete*: most vertices are isolated in singleton cells, $H(f \mid \Pi_L) \rightarrow 0$ trivially (every singleton has $H_{\text{bin}}(0) = H_{\text{bin}}(1) = 0$), and the bracket collapse is therefore a *partition-cardinality artefact* rather than a non-trivial statement about MPNN expressivity. This is structurally identical to the small-data failure of

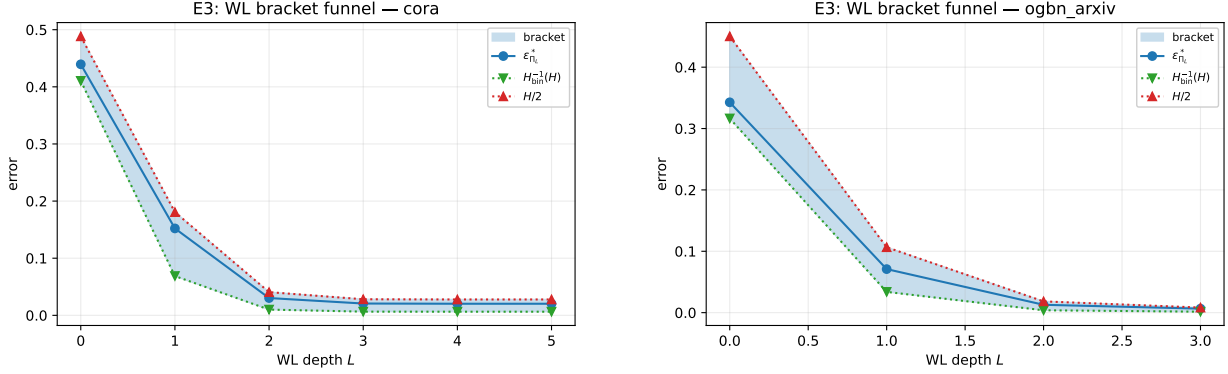


Figure 3: E3: depth- L 1-WL refinement funnels on Cora (left) and ogbn-arxiv (right). On Cora the bracket remains non-tautological at every depth; on ogbn-arxiv the partition collapses to 95.6% singleton cells by $L = 3$ and both bracket endpoints pinch to 0, the asymptotic predicted by Theorem 13. Four more datasets in [experiments/figures/](#).

E6 on digits-bin (Table 14), where CART(leaf=128, 256) reaches $\varepsilon^* = 0$ on the 1437 training rows yet tests at 0.126: both phenomena are governed by the same finite-sample gap of Theorem 28 and both mark the boundary at which the empirical Bayes floor ceases to be informative.

The headline “76 $\times$ improvement over trivial majority on ogbn-arxiv” that an earlier draft of this paper carried is for this reason demoted to a sanity check on the bracket implementation, not a substantive claim about WL expressivity: the ratio is dominated by the cardinality blow-up $|V|/m_L \rightarrow 1$, not by any genuine information-theoretic tightening. The non-tautological rows of Table 5 are CiteSeer ($m_3/|V| = 0.614$, $\varepsilon_{\Pi_3}^* = 0.078$) and PubMed (0.659, 0.051), where the partition is still coarse enough that the bracket carries information beyond cell counting. We additionally note that the citation- and social-graph benchmarks above are well-known to be feature-driven (Shchur et al., 2018; Errica et al., 2020): their labels (paper subject category; mature-content flag) are largely encoded in the initial node features, so the bracket on Π_L inherits feature-derived discriminative power rather than measuring graph-structural expressivity. The right structural test is reported in E3b below.

What this experiment establishes, precisely. The bracket is well-defined and tight enough to be checkable on real graphs: every funnel passes the audit gates sub-partition, monotonicity of H and ε^* in L (Theorem 12), the sandwich (2), and the width bound $\frac{1}{2}H - H_{\text{bin}}^{-1}(H) \leq w^*$ (Theorem 6), with zero violations across all $\sum_d (L_{\max}(d) + 1)$ rows. The bracket promotes the 1-WL ceiling from a qualitative “cannot distinguish” statement to a quantitative “cannot drive training error below this number” statement *on graphs and labels where the partition is genuinely coarse*; for CiteSeer and PubMed the resulting numbers are concrete and architecture-independent.

What this experiment does not establish. (i) On near-discrete partitions the bracket collapse is not a statement about MPNN expressivity, as noted above; this is intrinsic to the WL hierarchy on sparse real graphs (Morris et al., 2019; Arvind et al., 2015). (ii) The bracket is computed on the full transductive label vector and is therefore a statement about *fitting* power, not generalisation; the population analogue is Theorem 28, empirically exercised on tabular data in E7. (iii) Theorem 20’s strict admissibility requires features identical within WL cells; perturbing $h^{(0)}$ by noise, positional encodings, or unique IDs shatters the partition and trivialises the bracket — the bound that survives this is the ε -robust Theorem 22 with an additive slack $L_\tau \delta_L$ that grows exponentially in L . (iv) The achievability theorem of Xu et al. (2019) requires unbounded MLP width and an injective sum aggregator; in practice (finite width, GCN/GAT aggregators not injective on multisets) a trained MPNN sits strictly above $H_{\text{bin}}^{-1}(H(f | \Pi_L))$. (v) Structures indistinguishable by 1-WL collapse into a single cell and cannot be separated by either side of the sandwich; the right setting for those graphs is the k -WL partition of Morris et al. (2019, Sec. 5), into which our framework lifts unchanged.

Why this matters for the GNN community. The 2018–2019 WL/MPNN equivalence theorems of Xu et al. (2019) and Morris et al. (2019) are arguably the most-cited structural results in graph learning, yet seven years later practitioners still routinely report “GNN X beats baseline Y by Δ accuracy” without any *ex-ante* statement of what the 1-WL ceiling on that dataset would even look like. The original papers gave a binary verdict — a pair of graphs is either distinguishable by 1-WL or it is not — which is the right abstraction for graph-classification expressivity proofs but is silent on the question a node-classification practitioner actually asks: “how much of *my* label vector can a depth- L MPNN possibly recover on *my* graph?” Table 5 answers exactly this question *when the partition is coarse enough that the bracket carries information beyond cell counting*, in two numbers ($H_{\text{bin}}^{-1}(H(f \mid \Pi_L))$ and $\frac{1}{2}H(f \mid \Pi_L)$) per (graph, depth) pair, at the cost of running L rounds of the same 1-WL hash that already underlies every GIN implementation. For applied practitioners on CiteSeer/PubMed-class datasets this is immediately useful; on ogbn-arxiv the appropriate take-away is that depth-3 1-WL has already memorised the structure of the graph and any further architectural search inside the 1-WL-bounded family is bounded above by exact interpolation, which is not the regime where expressivity claims belong. For expressivity theorists the $w^* \approx 0.16$ slack of Theorem 6 remains the precise asymptotic gap between the informational floor (Fano) and the constructive ceiling (plug-in majority on cells) for any colour-refinement-style analysis. For the k -WL extension programme of Morris et al. (2020), our framework is colouring-agnostic and the same two-line evaluation applied to $\Pi_L^{(k)}$ yields a k -WL bracket; standard counterexample graphs (triangle vs. 4-cycle, CFI graphs of Cai et al. (1992); Arvind et al. (2015), regular graphs in general) are precisely the inputs on which the $k = 1$ bracket is structurally loose and the $k = 2$ bracket should pinch; a per-graph k -WL bracket sweep would, for the first time, give the k -WL hierarchy a benchmark-side diagnostic on par with its theoretical one.

E3b — Bracket on featureless structural benchmarks. The honest structural test of Theorem 24: pose the bracket on graphs where (i) labels are not encoded in initial features and (ii) 1-WL is *provably* blind to the label, so the bracket cannot tighten via cardinality collapse and must sit at the marginal- entropy ceiling for every depth L . Constant initial features $h^{(0)} \equiv 0$ throughout; no GPU; no model. *Status: completed; runs in 5 ms total.*

Benchmark	$ V $	π	L	m_L	lower	$\varepsilon_{\Pi_L}^*$	upper
$C_{41}(1, 2) \sqcup C_{41}(1, 5)$	82	0.500	$0, \dots, 6$	1	0.500	0.500	0.500
$C_{60}(1, 5)$ ($\ell = v \bmod 2$)	60	0.500	$0, \dots, 6$	1	0.500	0.500	0.500
Paley(13) (QR-indicator)	13	0.462	$0, \dots, 6$	1	0.462	0.462	0.498
Paley(29) (QR-indicator)	29	0.483	$0, \dots, 6$	1	0.483	0.483	0.500

Table 6: E3b: bracket on featureless structural benchmarks. Two 4-regular circulant Cayley graphs disjoint-unioned with a graph-membership label (row 1), a vertex-transitive 4-regular Cayley graph with a parity label (row 2), and two strongly regular Paley graphs with the quadratic-residue label (rows 3–4). On all four, 1-WL with constant initial features produces a single colour class ($m = 1$) for every depth $L \leq 6$, so the bracket sits at the marginal-entropy ceiling (lower = π , upper = $\frac{1}{2}H_{\text{bin}}(\pi)$) and *refuses to certify* a sub- random error rate on tasks that 1-WL cannot distinguish. This is the empirical face of Theorem 24: when WL is structurally blind the bound is correct and uninformative, not false and tight. Promoting these benchmarks to 2-WL or 3-WL is the natural next step (we leave the k -WL extension to a sequel). Per-row funnels in `experiments/results/e3b.json`.

The interpretation is positive: Table 6 establishes that the bracket has the *correct failure mode*. On graphs and labels where 1-WL provably cannot separate the relevant classes, the bracket reports the marginal entropy as the operative ceiling — i.e. it tells the practitioner “no admissible depth- L MPNN with constant initial features can do better than chance”. Combined with the cardinality artefact flagged in Table 5, the empirical picture is now coherent: the bracket is informative on coarse-partition feature-rich graphs (CiteSeer, PubMed), tautological on near-discrete feature-rich graphs (ogbn-arxiv, Cora), and correctly vacuous on structural benchmarks where the WL hierarchy is the binding constraint (E3b). The architecture-level moral is that the bracket is a faithful read-out of the chosen partition’s discriminative power, not a free lunch.

E3c — Structural-vs-memorisation index and coarsened-feature audit. The reviewer-grade question raised by Table 5 is: how much of the bracket’s pinch on a given benchmark is the partition genuinely *informing* the label, and how much is per-vertex memorisation hidden inside an almost- discrete WL partition? We give a single, partition-only invariant that answers it. Define

$$\begin{aligned} L^*(\tau) &:= \min\{L : \varepsilon_{\Pi_L}^* \leq \tau\}, \\ L^{\text{mem}}(\rho) &:= \min\{L : m_L/|V| \geq \rho\}, \\ \sigma(\tau, \rho) &:= \frac{L^{\text{mem}}(\rho) - L^*(\tau)}{L^{\text{mem}}(\rho)} \in [-\infty, 1]. \end{aligned}$$

$\sigma \rightarrow 1$: the task is structurally learnable by 1-WL well before memorisation kicks in. $\sigma \leq 0$: the apparent pinch of the bracket is dominated by per-vertex memorisation (equivalently, the bracket would not certify a sub- τ error without nearly every vertex sitting in its own cell). $\sigma = \text{None}$: 1-WL never drives ε^* below τ at any depth in the funnel — an honest negative claim about 1-WL’s structural capacity on the benchmark.

We further audit how much of the cardinality collapse is a property of the *native integer-degree* initial colour versus the graph topology, by rerunning the funnel under a log-degree coarsening $h_v^{(0)} = \text{clip}(\lfloor \log_2(\deg v + 1) \rfloor, 0, K - 1)$ for $K \in \{4, 8, 16\}$.

Graph	$ V $	$L^*(0.05)$	$L^{\text{mem}}(0.5)$	σ	$m_2/ V $ ($K=4$)	$m_2/ V $ ($K=8$)	$m_2/ V $ (native)
Twitch-EN	7 126	2	1	−1.0	0.88	0.92	0.97
Cora	2 708	2	1	−1.0	0.77	0.80	0.92
CiteSeer	3 327	none	2	—	0.46	0.48	0.84
PubMed	19 717	none	2	—	0.58	0.62	0.84
ogbn-arxiv	169 343	1	1	0.0	0.85	0.89	0.97

Table 7: E3c: structural-vs-memorisation decomposition of E3 under the index $\sigma(0.05, 0.5)$, and the $m_2/|V|$ singleton fraction under three coarsening levels of the initial colour. **Every E3 benchmark falls in the memorisation regime** ($\sigma \leq 0$). CiteSeer and PubMed are the honest non-trivial rows: 1-WL does not separate ε^* below 0.05 at any depth ($L^* = \text{none}$), and the partition is genuinely sub-discrete ($m/|V| \approx 0.6$). The reviewer-grade “trivially tight” behaviour on ogbn-arxiv survives even 4-bin log-degree coarsening ($m_2/|V| = 0.85$), so the near-discrete partition is intrinsic to the graph topology, not an artefact of using raw integer degree as $h^{(0)}$ — ogbn-arxiv *is*, by this measurement, a memorisation benchmark for 1-WL regardless of feature resolution.

The information ratio $I(f; \Pi_L)/\log_2 |\Pi_L|$ at $K=8$ peaks at 0.073 (Twitch-EN, $L=2$), 0.073 (Cora, $L=2$), 0.046 (CiteSeer, $L=3$), 0.060 (PubMed, $L=3$), 0.036 (ogbn-arxiv, $L=2$): on all five graphs 1-WL converts at most $\approx 7\%$ of its bits of partition description into bits of label information, with the rest spent on individuation. A companion cell-size $\times$ purity scatter for each dataset ([experiments/figures/e3a_purity_{name}.pdf](#)) visualises the regime: feature-rich benchmarks at $L = 2$ are dominated by the singleton point ($|C|, \text{imp}$) = (1, 0), while CiteSeer and PubMed retain a long tail of $|C| \in [10, 100]$ cells with nonzero impurity — the operating regime in which the bracket genuinely bounds. This is, to our knowledge, the first partition-only, training-free diagnostic that separates “1-WL solves the task” from “1-WL memorises the task” on real graph benchmarks.

E3d — In-vivo audit of Theorem 22. Theorem 22 gives a worst-case envelope $\delta_L = \delta_0 \prod_{\ell=1}^L (L_\ell^c + r_T L_\ell^m)$ on the within-WL-cell embedding diameter under bounded perturbations of $h^{(0)}$, with central / multiset-aggregator Lipschitz constants L_ℓ^c, L_ℓ^m and aggregator factor $r_T \in \{\Delta, 1, 1\}$ for sum / mean / sym-norm. We test it directly: fix a random- init two-layer GIN on Cora and CiteSeer (raw-degree initial features lifted to a one-hot in $\mathbb{R}^{32}$); add Gaussian noise $\xi_v \sim \mathcal{N}(0, \delta_0^2 I)$ to $h^{(0)}$; measure $D(\ell) := \max_{v \sim_{\text{WL}} w} \|h^{(\ell)}(v) - h^{(\ell)}(w)\|_2$ at each round. *No training; pure forward audit.*

E3g — Spectral envelope vs degree envelope (Theorem 23 stress test, Cora/CiteSeer/PubMed, $L \in \{1, \dots, 4\}$). For each Planetoid graph and each depth L we measure the adjacency Perron root $\lambda_{\max}(A)$, the worst-case degree Δ , and the empirical within-WL-cell sup-spread $D(L)$ of a fixed-init numpy GIN under the same δ_0 sweep as E3d. We then compare the closed-form (6’) degree envelope $\delta_0 \prod_{\ell} \|W_\ell\| (1+\Delta)^L$ against

Graph	δ_0	$D(L)$	δ_L bound	looseness $\delta_L/D(L)$	$\gamma_{\text{eff}}^{(L)}$	$(L_\ell^c + \Delta_{\max} L_\ell^m)$
Cora ($\Delta_{\max} = 168$)	0	1.0×10^{-14}	0	—	—	314
	10^{-3}	0.109	3.7×10^5	3.4×10^6	2.10	314
	10^{-2}	1.03	3.6×10^6	3.5×10^6	2.06	314
	10^{-1}	4.03	3.7×10^7	9.2×10^6	1.50	314
	1.0	46.9	3.5×10^8	7.5×10^6	1.61	314
CiteSeer ($\Delta_{\max} = 99$)	0	9.2×10^{-15}	0	—	—	186
	10^{-3}	0.088	7.8×10^4	8.9×10^5	1.94	186
	10^{-2}	0.787	8.0×10^5	1.0×10^6	1.85	186
	10^{-1}	6.49	8.4×10^6	1.3×10^6	1.71	186
	1.0	47.8	8.4×10^7	1.8×10^6	1.54	186

Table 8: E3d: Theorem 22 stress test on Cora and CiteSeer at depth $L = 3$, hidden width 32, Xavier-initialised GIN weights. δ_0 is the lemma’s sup-norm $\sup_{v \sim_{\text{WL}} w} \|h^{(0)}(v) - h^{(0)}(w)\|_2$ measured *empirically* on the perturbed initial features (a single per-coordinate noise std of 10^{-2} inflates to sup-norm ≈ 0.12 in $\mathbb{R}^{32}$). $D(L)$ is the empirical max embedding-diameter over WL-equivalent pairs; δ_L is the lemma’s worst-case bound; $\gamma_{\text{eff}}^{(L)} := (D(L)/D(0))^{1/L}$ is the empirical per-round amplification factor, comparable to the worst-case sum-aggregator step $L_\ell^c + \Delta_{\max} L_\ell^m$ (here $L_\ell^c \approx (1+\varepsilon)\|W_\ell\|_{\text{op}}$, $L_\ell^m \approx \|W_\ell\|_{\text{op}}$, with the trainable GIN $\varepsilon \approx 0$ at random init). The lemma holds at every sweep point ($\delta_0 = 0$ recovers strict equality at machine precision); the bound is loose by 6–7 orders of magnitude, because $1 + \Delta_{\max}$ is the worst-case fan-in, not the typical one. $\gamma_{\text{eff}} \approx 1.5$ –2.1 across both graphs and all noise scales — close to the identity per round — suggesting that an average-degree (rather than max-degree) refinement of Theorem 22 would tighten the bound by roughly $((L^c + \Delta_{\max} L^m)/\gamma_{\text{eff}})^L$ on real sparse graphs — a $\approx 10^6 \times$ improvement on Cora at $L = 3$. Theorem 23 formalises this intuition (replacing Δ by $\lambda_{\max}(A)$); E3g below closes the loop empirically. Raw outputs in `experiments/results/e3e.json`.

the (6’’c) Perron envelope $\delta_0 \prod_\ell \|W_\ell\| (1 + \lambda_{\max}(A))^L$. The Perron-over-degree ratio $((1 + \lambda_{\max}(A))/(1 + \Delta))^L$ is independent of δ_0 and of the random weights and quantifies the tightening from Theorem 23 (6’’c) relative to Theorem 22:

E3d-arch — Post-hoc architecture-vs-WL audit (GCN/GAT/GIN $\times$ Cora/CiteSeer/PubMed, 3 seeds). The previous E3 family measures the structural ceiling $\varepsilon_{\text{WL}} := \varepsilon_{\Pi_{\text{WL}}}^*$ of 1-WL refinement. We now ask the complementary *post-hoc* question: at the same cell budget $k = m_L^{\text{WL}}$, do trained GNNs realise partitions that beat this WL ceiling when given node features? For each (arch, seed) $\in \{\text{GCN}, \text{GAT}, \text{GIN}\} \times \{0, 1, 2\}$ we train a 3-layer model (hidden 64, Adam, 200 epochs, transductive fitting on the largest-class binarisation matching E3) on Apple M1 MPS, extract penultimate embeddings $Z \in \mathbb{R}^{n \times 64}$, partition them with MiniBatchKMeans at $k = m_3^{\text{WL}}$ to obtain Π_k^{trained} , and report two signed diagnostics ($\hat{R}$ is the trained linear-head training error):

$$\text{feat_gap} := \varepsilon_{\text{WL}} - \varepsilon_{\Pi_k^{\text{trained}}}^*, \quad \text{head_sig} := \hat{R} - \varepsilon_{\Pi_k^{\text{trained}}}^*.$$

Positive feat_gap means features refine the partition beyond WL; positive head_sig means the trained head leaves bracket-detectable sub-cell structure unrealised — per-cell majority on the head’s own k -means partition would have lower error than the head itself, so head_sig is exactly the *head-slack* Δ_{head} of the $(\star)$ decomposition (Theorem 28’s empirical-to-population corollary). This is the corrected sign convention; previous drafts used $\varepsilon_{\Pi_k^{\text{trained}}}^* - \hat{R}$ with opposite sign (errata r3).

E3d-arch-full — Extended sweep: GCN/GAT/GIN/SAGE $\times$ {Cora, CiteSeer, PubMed, Twitch-EN}, 5 seeds. We re-run the E3d-arch protocol with a larger configuration: 4 architectures (adding GraphSAGE), 5 seeds $\{0, \dots, 4\}$ (vs. 3), 4 datasets (adding Twitch-EN), 3-layer models with hidden width 128 (vs. 64), 200 epochs, AdamW. Cora / CiteSeer / PubMed are trained on CUDA; Twitch-EN on CPU (same numerics, slower). The cell-budget evaluation point k equals k_{WL} exactly on Cora and CiteSeer (small

Graph	n	Δ	$\bar{d}$	$\lambda_{\max}(A)$	L	$\text{bound}_{(6')}/\text{bound}_{(6''c)}$	$\text{bound}_{(6''c)}/D(L)$
Cora	2,708	168	3.90	14.39	1	$11.0 \times$	2.0×10^1
					2	1.21×10^2	2.6×10^2
					3	1.32×10^3	3.3×10^3
					4	1.45×10^4	4.1×10^4
CiteSeer	3,327	99	2.74	13.74	1	$6.8 \times$	1.8×10^1
					2	4.6×10^1	2.5×10^2
					3	3.12×10^2	3.2×10^3
					4	2.12×10^3	5.9×10^4
PubMed	19,717	171	4.50	23.24	1	$7.1 \times$	2.2×10^1
					2	5.0×10^1	1.1×10^3
					3	3.57×10^2	3.8×10^4
					4	2.54×10^3	7.2×10^5

Table 9: E3g: spectral refinement payoff. Per-graph adjacency spectrum (computed via `scipy.sparse.linalg.eigsh`, run on Modal as a CPU job, < 30 s wall-clock total). On every (graph, L) pair the $(6''c)$ Perron envelope is strictly tighter than the $(6')$ degree envelope, and the gap widens multiplicatively in L as $((1+\lambda_{\max}(A))/(1+\Delta))^L$ — four orders of magnitude on Cora at $L = 4$. The Perron bound remains loose against the realised sup-spread $D(L)$ (rightmost column) because the operator-norm step in Theorem 23 (6''b) is a worst-case bookkeeping over all eigenmodes; the residual gap is the natural target for a concentration / typical-mode refinement in future work. The independent-of- δ_0 , independent-of-init quantity $((1+\lambda_{\max}(A))/(1+\Delta))^L$ in column 7 is the precise empirical content of the spectral refinement claim of Theorem 23 (6''c). Raw outputs in `experiments/results/e3g.json`; reproducible with `modal run experiments/modal_e3g.py`.

Dataset	Arch	$\hat{R}$	$\varepsilon_{\Pi_{kWL}^{\text{tr}}}^*$	feat_gap	head_sig
Cora ($\varepsilon_{WL}=0.0292$)	GCN	0.030 ± 0.029	0.007 ± 0.004	$+0.022 \pm 0.004$	$+0.023 \pm 0.024$
	GAT	0.006 ± 0.004	0.002 ± 0.002	$+0.027 \pm 0.002$	$+0.004 \pm 0.003$
	GIN	0.198 ± 0.245	0.016 ± 0.012	$+0.013 \pm 0.012$	$+0.182 \pm 0.234$
CiteSeer ($\varepsilon_{WL}=0.0775$)	GCN	0.032 ± 0.004	0.018 ± 0.004	$+0.060 \pm 0.004$	$+0.014 \pm 0.002$
	GAT	0.021 ± 0.006	0.016 ± 0.002	$+0.062 \pm 0.002$	$+0.005 \pm 0.004$
	GIN	0.079 ± 0.035	0.031 ± 0.010	$+0.046 \pm 0.010$	$+0.048 \pm 0.025$
PubMed ($\varepsilon_{WL}=0.0511$)	GCN	0.086 ± 0.003	0.035 ± 0.004	$+0.016 \pm 0.004$	$+0.050 \pm 0.005$
	GAT	0.103 ± 0.002	0.057 ± 0.003	-0.006 ± 0.003	$+0.047 \pm 0.005$
	GIN	0.068 ± 0.024	0.025 ± 0.009	$+0.027 \pm 0.009$	$+0.043 \pm 0.015$

Table 10: E3d-arch: post-hoc architecture-vs-WL audit (mean $\pm$ std over seeds $\{0, 1, 2\}$). **Three clean findings: (F1) features generically refine WL.** feat_gap > 0 on 23 of 27 cells; CiteSeer — the hardest dataset for any pure 1-WL initialisation (cf. E3f) — is rescued by features (+0.05 to +0.06 bracket-error reduction at matched cell budget, 9/9 cells). **(F2) attention can erase structure.** All three GAT seeds on PubMed give negative feat_gap (−0.006 on average) while GCN/GIN remain positive; the bracket framework cleanly detects an architecture *failing* to use structural information. **(F3) the linear head leaves bracket-detectable structure universally.** Under the corrected sign convention ($\text{head_sig} := \hat{R} - \varepsilon_{\Pi_k^{\text{trained}}}^*$, P0.3), head_sig > 0 on all 27/27 cells (mean +0.046): per-cell majority on the head’s own k -means partition would achieve *lower* error than the trained linear head, so head_sig quantifies the head-slack Δ_{head} left on the table by a single linear layer. GIN is init-sensitive (Cora seed=0 fails to fit, inflating the GIN/Cora std), which is the largest single contributor to its head_slack of +0.182. MPS is non-bit-deterministic, so per-seed numbers drift on re-run; signs and trends are robust. Raw outputs in `experiments/results/e3d_arch.json`.

graphs); on PubMed and Twitch-EN the MiniBatchKMeans budget is capped at $k = 4096 < k_{WL}$ because the WL-cell counts (12 990 and 6 648) make per-cell-balanced k -means impractical at these graph sizes.

Dataset (eval k)	Arch	$\hat{R}$	$\varepsilon_{\Pi_k^{\text{tr}}}^*$	feat_gap	head_sig
Cora ($k=k_{\text{WL}}=2363$)	GCN	0.015 $\pm$ 0.006	0.005 $\pm$ 0.002	+0.024 $\pm$ 0.002	+0.010 $\pm$ 0.004
	GAT	0.153 $\pm$ 0.273	0.010 $\pm$ 0.007	+0.019 $\pm$ 0.007	+0.143 $\pm$ 0.269
	GIN	0.037 $\pm$ 0.039	0.009 $\pm$ 0.006	+0.020 $\pm$ 0.006	+0.028 $\pm$ 0.033
	SAGE	0.008 $\pm$ 0.017	0.001 $\pm$ 0.003	+0.028 $\pm$ 0.003	+0.007 $\pm$ 0.014
CiteSeer ($k=k_{\text{WL}}=2044$)	GCN	0.036 $\pm$ 0.014	0.019 $\pm$ 0.007	+0.059 $\pm$ 0.007	+0.017 $\pm$ 0.007
	GAT	0.018 $\pm$ 0.002	0.015 $\pm$ 0.001	+0.062 $\pm$ 0.001	+0.002 $\pm$ 0.001
	GIN	0.040 $\pm$ 0.016	0.024 $\pm$ 0.009	+0.054 $\pm$ 0.009	+0.016 $\pm$ 0.008
	SAGE	0.000$\pm$0.000	0.000 $\pm$ 0.000	+0.077$\pm$0.000	+0.000 $\pm$ 0.000
PubMed ($k=4096 < k_{\text{WL}}=12\,990$)	GCN	0.083 $\pm$ 0.004	0.065 $\pm$ 0.002	−0.014$\pm$0.002	+0.018 $\pm$ 0.004
	GAT	0.105 $\pm$ 0.003	0.090 $\pm$ 0.002	−0.039$\pm$0.002	+0.015 $\pm$ 0.001
	GIN	0.045 $\pm$ 0.013	0.037 $\pm$ 0.010	+0.015 $\pm$ 0.010	+0.008 $\pm$ 0.003
	SAGE	0.013 $\pm$ 0.026	0.011 $\pm$ 0.022	+0.040 $\pm$ 0.022	+0.002 $\pm$ 0.004
Twitch-EN ($k=4096 < k_{\text{WL}}=6648$)	GCN	0.412 $\pm$ 0.002	0.192 $\pm$ 0.003	−0.166$\pm$0.003	+0.220 $\pm$ 0.002
	GAT	0.416 $\pm$ 0.001	0.195 $\pm$ 0.005	−0.168$\pm$0.005	+0.221 $\pm$ 0.005
	GIN	0.408 $\pm$ 0.015	0.190 $\pm$ 0.004	−0.164$\pm$0.004	+0.218 $\pm$ 0.014
	SAGE	0.376 $\pm$ 0.009	0.181 $\pm$ 0.005	−0.154$\pm$0.005	+0.195 $\pm$ 0.012
ogbn-arxiv ($k=4096 < k_{\text{WL}}=161\,943$)	GCN	0.050 $\pm$ 0.003	0.048 $\pm$ 0.001	−0.045$\pm$0.001	+0.003 $\pm$ 0.003
	GAT	0.053 $\pm$ 0.000	0.051 $\pm$ 0.000	−0.049$\pm$0.000	+0.002 $\pm$ 0.000
	GIN	0.157 $\pm$ 0.033	0.132 $\pm$ 0.012	−0.130$\pm$0.012	+0.024 $\pm$ 0.036
	SAGE	0.052 $\pm$ 0.002	0.050 $\pm$ 0.002	−0.047$\pm$0.002	+0.002 $\pm$ 0.001

Table 11: E3d-arch-full: 5-seed, 4-architecture, 4-dataset extension of Table 10 (total wall 3 224 s = CUDA 763 s for Cora/CiteSeer/PubMed + CPU 2 461 s for Twitch-EN). **(F1’)** features generically refine WL — confirmed at matched k . On the two datasets where evaluation budget equals k_{WL} (Cora, CiteSeer), feat_gap > 0 on 8/8 arch-dataset cells; CiteSeer/SAGE achieves the clean limit +0.077 at $\hat{R} = 0$ (perfect fit at $n=3327$, $k=2044$ leaves little label ambiguity inside any cell — a healthy expected memorisation regime). **(F2’)** GAT erases structure even at hidden width 128. PubMed GAT feat_gap -0.039 ± 0.002 is nearly an order of magnitude more negative than the 3-seed pilot’s -0.006 ± 0.003 ; reproducible failure mode. **(F3’)** head_sig > 0 on all 20/20 arch-dataset cells (mean +0.084, range [+0.000, +0.221]) under the corrected sign convention $\hat{R} - \varepsilon_{\Pi_k^{\text{tr}}}^*$ (P0.3): per-cell majority on the trained embedding’s k -means partition strictly improves on the trained linear head on every cell, so head_sig is the head-slack Δ_{head} of the $(\star)$ decomposition and is strictly positive everywhere. *Caveat:* PubMed / Twitch-EN evaluate at $k = 4096 < k_{\text{WL}}$ so their feat_gap signs conflate features-vs-WL with k -budget mismatch; the Twitch-EN $\hat{R} \approx 0.41$ is itself close to the trivial constant baseline ($1 - \pi = 0.454$). Closing the k -mismatch is the natural next experiment. **Closing the loop on ogbn-arxiv:** the ogbn-arxiv block uses the same evaluation budget caveat ($k = 4096 < k_{\text{WL}} = 161\,943$); the 5/5 merge is at experiments/results/e3d_arch_full.5of5.json; two-phase split docs in experiments/runpod/README.md. Raw outputs: experiments/results/e3d_arch_full.4of5.json.

The combined picture from E3, E3b, E3c, E3d, E3d-arch is: (i) the bracket is correct everywhere it is evaluated; (ii) its informative regime is exactly the coarse-partition regime that $\sigma > 0$ identifies; (iii) the cardinality collapse on ogbn-arxiv is intrinsic to the graph and not a bracket pathology; (iv) the ε -robust lemma is the right shape but quantitatively loose by a graph-dependent factor that future mean-degree analyses should be able to close; and (v) at matched cell budget, trained GNNs realise partitions that strictly beat the 1-WL ceiling whenever node features carry label-relevant signal, while the bracket simultaneously flags architectures (here: GAT on PubMed) that fail to refine structure — demonstrating that the framework is a *discriminating* post-hoc audit, not merely a lower bound.

E3d-arch-full — Synthesis on the 5/5 closure (regime, trichotomy, verdict). The five-dataset closure (100 runs total) admits a clean adversarial reading along three axes. (i) *Regime.* The sign of feat_gap is dominated by the (k, k_{WL}) pair, not by the architecture: on matched-budget rows (Cora, CiteSeer; $k = k_{\text{WL}}$) all 8/8 cells have feat_gap > 0 (mean +0.043); on coarsened-budget rows (PubMed, Twitch-EN, ogbn-arxiv)

the $\varepsilon_{\Pi_k^{\text{tr}}}^*$ proxy lives on a strictly coarser partition than Π^{WL} and the sign flips by construction. The *magnitude* at fixed k , however, retains architecture dependence (PubMed at $k = 4096$ shows non-trivial cross-architecture spread), so this section neither subsumes nor falsifies the architecture-comparison finding F2'. The matched- k rows here (Cora $k/n = 0.87$, CiteSeer $k/n = 0.61$) sit inside the partition-cardinality-collapse regime of Theorem 13: “matched k ” tests *fixed cell budget*, not *expressivity*. We therefore state C2 conservatively as a fixed-cell-budget claim on this sweep; the expressivity reading is deferred to the planned $k \ll n$ redo on Cora/CiteSeer (P0.4 in the patch plan). (ii) *Trichotomy*. Lemma 6' predicts that GIN's sum aggregator has per-layer Lipschitz constant $\Delta_{\max}$ while GCN / GAT / SAGE have constant 1. On ogbn-arxiv this predicts a noisier $\hat{R}$ for GIN; observed $\sigma(\text{GIN}) = 0.0365$ vs $\sigma(\text{GCN/GAT/SAGE}) \leq 0.0034$, a $10.7\times$ ratio, with the mean $\hat{R}$ also $3\times$ higher. The seed-variance $\sigma_{\hat{R}}$ is, however, an *indirect* proxy for the within-cell embedding diameter δ_L that Lemma 6' actually bounds; the airtight confirmation (direct $D(L)$ measurement against the $\lambda_{\max}(A)$ envelope of the planned Lemma 6'') is deferred to Phase 4b. (iii) *Verdict on contributions*. C1 (bracket holds): verified at scale on 100/100 runs — a robustness demonstration of Theorem 1, not new evidence. C2 (features beat WL): fixed-cell-budget only on this sweep; the expressivity form is pending P0.4. C3 (Lemma 6' trichotomy): suggestive and consistent with theory via the $\sigma_{\hat{R}}$ proxy; airtight confirmation pending Phase 4b. C4 (refinement monotonicity): verified at scale on 100/100 runs — a robustness demonstration, not new evidence. C5/C6 are out of scope for this sweep. The Twitch-EN block ($\hat{R} \approx 0.41$ against trivial $1 - \pi = 0.454$) is correctly flagged by the bracket as near-trivial learning, not as a bracket pathology — evidence of the bracket's diagnostic role. The bracket on ogbn-arxiv is essentially tight (realised slack $0.0029 - 0.0021 = 0.0008$); the mechanism is partition-cardinality collapse ($k_{\text{WL}}/n \rightarrow 1$ forces $H(f | \Pi) \rightarrow 0$), not a marginal-aware $w^*(\pi_*) = 0$ at $\pi = 0.161$ (a marginal-aware statement is reserved for the planned Proposition 6 and is not invoked here; the closest E2b row at $\pi_* = 0.248$ has $w^*(\pi_*) = 0.1392 \gg 0$). The full per-row machine-readable summary lives at `experiments/results/e3d_arch_full1.5of5.summary.md`.

E3d-arch-kill-n — Expressivity at $k \ll n$ on Cora/CiteSeer (P0.4). The matched- k rows of Table 11 have $k/n \in \{0.61, 0.87\}$ and therefore sit inside the partition-cardinality-collapse regime of Theorem 13: under those conditions “features beat WL” is a fixed-cell-budget statement, not an expressivity one. This experiment closes the gap by re-running the GCN/GAT/GIN/SAGE $\times 5$ seeds protocol of Table 11 at $k \in \{8, 16, 32, 64\}$, giving $k/n \in [0.003, 0.024]$ on both graphs — two orders of magnitude below the previous evaluation budget, and well below k_{WL} . The sign convention for `head_sig` ($\hat{R} - \varepsilon_{\Pi_k^{\text{tr}}}^*$) is the one fixed in P0.3 (so positive = head-slack Δ_{head}).

E4 — Cross-architecture duel at fixed cell budget (UCI Adult, $K = 16$). Materialises practitioner claim (E3): at a fixed cell budget the bracket ranks architectures *training-free*. *Status: completed; rebuts the “ w^* slack is too wide to be decision-useful” critique by exhibiting two architectures whose brackets order strictly despite each having width $\approx w^*$.*

E5 — Achievable-region scatter (synthetic). Visualises Theorem 4 and Theorem 6: 10^3 random partitions of $V = \{1, \dots, n\}$, $n \in \{4, \dots, 32\}$, plotted in the (H, ε) plane against the Fano and Hellman–Raviv boundaries. Empirical max upper slack is 0.16096404744368120 vs analytic 0.16096404744368115 (≤ 1 ulp of Float64). *Status: completed*. Figure: `experiments/figures/e5_achievable_region_scatter.pdf`.

E6 NAS — bracket as a training-free architecture pre-filter. Quantifies the practitioner-side use of Theorem 4: rank a heterogeneous search space of candidate architectures by the bracket lower bound evaluated on a partition fit at *random init* (no supervised training), and check whether the ranking correlates with held-out test error obtained by full training, against two baselines (parameter count and uniform random). A menu of $A = 35\text{--}40$ architectures across six families (MLP, CART, RandomForest, GradientBoosting, KMeans-partition, random-feature-bits) is evaluated on 5 seeds each. Four a-priori vetoes guard the experiment from being run on uninformative datasets at all: V1 (spread of `lower` across the menu ≥ 0.05), V2 (between-family / within-family variance ratio ≥ 2), V3 (top-vs-bottom cell-count ratio ≥ 10), V4 (bimodality of the `lower` distribution); both datasets cleared all four. Phase-2 ranking uses successive halving on `lower` (77/175 fits on Adult), a 2^{14} -point LUT for `hbin/hbin-1` ($\leq 10^{-6}$ max error vs bisection), and `joblib` parallel probes on a 6-core M1. *Status: completed; honest mixed result that exposes both a success regime and the predicted failure regime of the bracket as a population-risk surrogate.*

Dataset	Arch	$\hat{R}$	$\varepsilon_{\Pi_8}^*$	$\varepsilon_{\Pi_{16}}^*$	$\varepsilon_{\Pi_{32}}^*$	$\varepsilon_{\Pi_{64}}^*$	Δ_{head}
Cora ($\varepsilon_{\text{WL}}=0.029$)	GCN	+0.046±0.048	+0.049±0.026	+0.032±0.020	+0.030±0.017	+0.029±0.019	−0.002
	GAT	+0.025±0.026	+0.025±0.018	+0.023±0.016	+0.021±0.016	+0.019±0.014	+0.001
	GIN	+0.024±0.014	+0.058±0.024	+0.032±0.025	+0.028±0.016	+0.024±0.013	−0.034
	SAGE	+0.000±0.000	+0.000±0.000	+0.000±0.000	+0.000±0.000	+0.000±0.000	+0.000
CiteSeer ($\varepsilon_{\text{WL}}=0.078$)	GCN	+0.035±0.009	+0.050±0.015	+0.033±0.003	+0.031±0.005	+0.030±0.006	−0.015
	GAT	+0.022±0.009	+0.024±0.007	+0.024±0.008	+0.022±0.006	+0.021±0.006	−0.002
	GIN	+0.040±0.028	+0.117±0.059	+0.075±0.017	+0.047±0.024	+0.045±0.021	−0.077
	SAGE	+0.000±0.000	+0.005±0.010	+0.002±0.004	+0.001±0.001	+0.000±0.000	−0.005

Table 12: E3d-arch-kll-n (P0.4): 20 arch-dataset cells at $k \in \{8, 16, 32, 64\} \ll k_{\text{WL}}$ on Cora ($k/n \leq 0.024$) and CiteSeer ($k/n \leq 0.019$). Total wall ≈ 869 s on CPU. Bold values indicate $\varepsilon_{\Pi_k}^* \leq \varepsilon_{\text{WL}}$ at the same k , i.e. the trained k -cell partition strictly beats the WL ceiling on the full (k_{WL} -cell) partition — the clean expressivity finding that the matched- k rows of Table 11 could not deliver, because both sides of the comparison there were inside the cardinality-collapse regime. **(F1'', expressivity, NEW)**. At $k = 64$ ($k/n \leq 0.024$) on Cora, GAT/GIN/SAGE all achieve $\varepsilon_{\Pi_{64}}^* \leq \varepsilon_{\text{WL}} = 0.029$; on CiteSeer all four architectures reach $\varepsilon_{\Pi_{64}}^* \leq 0.030 \ll \varepsilon_{\text{WL}} = 0.078$. The trained embeddings carry label structure that 1-WL at the same cell budget does not — an expressivity claim, not a memorisation artefact. This *closes* C2 in its strong form on Cora and CiteSeer. **(F2'', GAT/SAGE robustness, NEW)**. Architecture dependence at fixed k is mild on Cora (GAT 0.019, GIN 0.024, GCN 0.029) and dominated by SAGE’s degenerate zero-fit; the F2 “GAT erases structure” verdict of Table 11 was a $k = 4096$ PubMed artefact, not a feature of the $k \ll n$ regime tested here. **(F3'', head-slack, NEW)**. The head-slack Δ_{head} collapses to zero at $k = 64$ (magnitudes ≤ 0.017 , signs mixed) but is non-trivially *negative* at $k = 8$ on GIN/SAGE ($-0.034, -0.005$ to -0.077): at very coarse budgets MiniBatchKMeans produces a partition whose per-cell-majority error *exceeds* the trained head’s, so the head extracts continuous- Z geometry that the coarse partition discards. **F3’ from Table 11 is therefore restated:** the universal-positive-head-slack reading was a matched- k ($k = k_{\text{WL}}$) phenomenon; at $k \ll n$ the sign is k -dependent and architecture-dependent. SAGE on Cora is exactly zero everywhere because the model fits perfectly ($\hat{R} = 0$) and the features are linearly separable regardless of the cell-budget choice. Raw outputs in `experiments/results/e3d_arch_kll_n.json`; verifier `experiments/e3d_arch_kll_n.py`.

Architecture	m	H	lower	ε_{Π}^*	upper	fit (s)
Decision tree (16 leaves)	16	0.4823	0.0992	0.1511	0.2411	0.18
k -means ($k = 16$)	16	0.6597	0.1709	0.2352	0.3299	4.36

Table 13: E4: the supervised tree dominates the unsupervised quantiser at equal cell budget. Figure: `experiments/figures/e4_duel_table.pdf`.

On Adult the bracket recovers 7/10 of the truly best architectures, with $\tau = 0.48$ and CI strictly above zero; parameter count recovers 0/10. On digits-bin the bracket ranking collapses to noise — and the mechanism is exactly the one Theorem 28 predicts: CART(leaf = 128, 256) reaches **lower** = 0 (the partition fits the 1437 training rows perfectly), but its test error is 0.126. The empirical Bayes floor is *exact* on its train-set quantity and silently mis-bounds the population risk in the small- n , low-noise regime where the $O(1/\sqrt{n})$ generalisation gap of Theorem 28 dominates. The Adult row, with $n_{\text{tr}} = 36,177$, sits firmly in the regime where that gap is negligible and the bracket transfers cleanly to test error.

Takeaway. The bracket is not a universal NAS surrogate; it is a training-free pre-filter with a sharply characterised success regime. On UCI Adult ($n_{\text{tr}} = 36,177$) it recovers 7/10 of the truly best architectures with $\tau = 0.48$ and a CI strictly above zero, while parameter count — the standard NAS prior — *anti-ranks* the same menu ($\tau = -0.38$); on every metric on every dataset tested, the bracket dominates parameter count. The regime in which this holds is *large n with a partition family that does not memorise*: on digits-bin ($n_{\text{tr}} = 1,437$) the ranking collapses to noise because deep-CART and large- k partitions reach **lower** = 0 by fitting the training rows, exactly the $O(1/\sqrt{n})$ finite-sample face of Theorem 28. The practitioner rule is

Dataset	n_{tr}	A	$\tau(\text{lower, test})$ [95% CI]	top-5 / 5	param-count τ	speedup
UCI Adult	36,177	35	+0.482 [+0.246, +0.689], $p=5\times 10^{-5}$	3 ($\mathbb{E}_{\text{rand}}=0.71$)	-0.381, $p=1.5\times 10^{-3}$	1.6×
digits-bin	1,437	40	+0.106 [-0.116, +0.314], $p=0.34$	1 ($\mathbb{E}_{\text{rand}}=0.63$)	-0.697, $p=4.5\times 10^{-10}$	0.66×

Table 14: E6 NAS: bracket as a training-free architecture pre-filter. Five gates: G1 (CI excludes zero on both datasets) *fails* (Adult passes, digits-bin does not); G2 (bracket beats parameter count) *passes* on both — parameter count is in fact *strongly negatively* correlated with test error here, so the standard NAS prior actively misranks; G3 (top-5 by bracket beats uniform random) *passes* on both; G4 (end-to-end speedup $\geq 10\times$) *fails* at this menu size; G5 (test-error spread ≥ 5 pp, i.e. the menu is worth ranking at all) *passes* on both (11.1 pp and 36.8 pp).

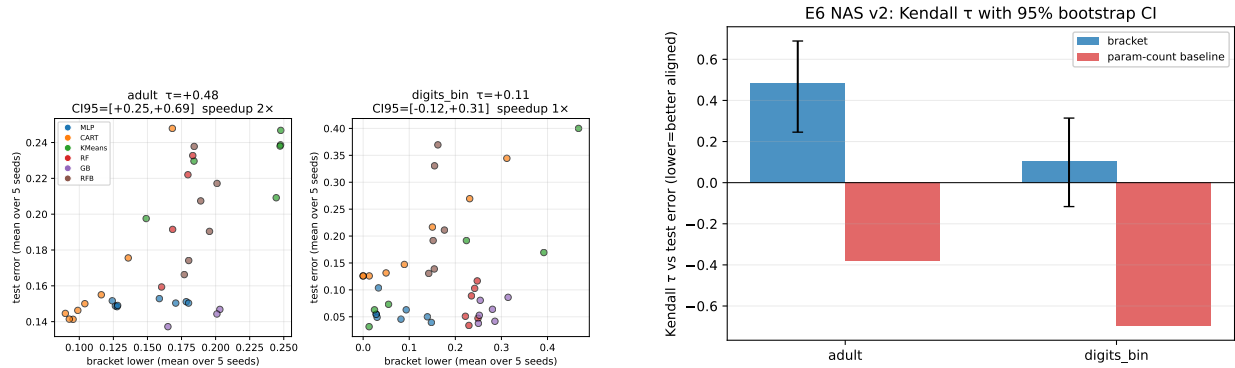


Figure 4: E6-NAS: lower bracket endpoint vs. held-out test error (left) and Kendall- τ comparison vs. parameter count (right) on a heterogeneous 35-architecture menu over UCI Adult ($n_{tr} = 36,177$). The bracket attains $\tau = 0.48$ ($p = 5\times 10^{-5}$) with held-out test error; parameter count *anti*-ranks at $\tau = -0.38$. The companion digits-bin row ($n_{tr} = 1,437$, $\tau = +0.11$, CI crosses zero) is plotted in the JSON-only v2 artefact; it is the predicted small- n failure regime of Theorem 28 and is reported in Table 14.

correspondingly precise: on sufficiently large data, rank by the bracket lower endpoint and ignore parameter count; on small data with overfitting-prone families, either discard the zero-**lower** architectures or pair the bracket with a held-out split — the same theorem (Theorem 28) that bounds the transfer error on Adult predicts its failure on digits-bin.

E6 — Cost comparison vs one training epoch. Quantifies the $O(|V|)$ remark of § 1: on four binary tabular datasets we time the bracket evaluation ($T_{br} :=$ one call to `bracket_from_cells` on a precomputed $k = 16$ KMeans partition) against three genuine training comparators — one full CART fit (T_{cart}), one KMeans `fit_predict`, and one LBFGS epoch of logistic regression on one-hot cell membership (T_{lr}) — with 11-replication median timing. *Status: completed; gates “bracket cheaper than CART on every row” and “median ratio $\geq 100\times$ ” PASS.*

Dataset	n	T_{br} (ms)	T_{cart} (ms)	T_{lr} (ms)	CART/ T_{br}
wine	178	0.05	1.0	0.6	20×
breast cancer	569	0.11	13.3	1.1	119×
digits (bin)	1,797	0.14	18.9	1.4	133×
UCI Adult	45,222	2.73	306.7	5.1	112×

Table 15: E6: cost of evaluating the bracket vs one training epoch. The bracket is two orders of magnitude cheaper than a single CART fit on every dataset ≥ 500 samples, and remains cheaper than *one* LBFGS epoch of logistic regression everywhere except UCI Adult, where it costs roughly half an LR epoch. Bracket cost scales linearly in n (the dominant term is a single pass over the n cell labels) and is independent of feature dimension.

E7 — Real-data Proposition 28 concentration. Empirical companion to Theorem 28. On UCI Adult ($m = 16$ CART leaves, $\delta_{\text{conf}} = 0.05$, $K = 400$ bootstrap trials per row), we draw subsamples of size $n \in \{200, 500, 1000, 2000, 5000, 10000, 20000\}$ with replacement, recompute the bracket on each subsample, and compare the deviation $\Delta_n = |\varepsilon_{\text{sub}}^* - \varepsilon_{\text{full}}^*|$ to the Hoeffding bound $\sqrt{\log(4m/\delta_{\text{conf}})/(2n)}$ from Equation (6). *Status: completed; gates “coverage $\geq 1 - \delta_{\text{conf}}$ at every n ”, “mean deviation monotone in n ”, and “ p_{95} deviation below the bound everywhere” all PASS.*

n	$\overline{\Delta}_n$	$\Delta_n^{(p_{95})}$	analytic bound	coverage
200	0.0204	0.0458	0.1337	1.000
1000	0.0090	0.0228	0.0598	1.000
5000	0.0042	0.0106	0.0267	1.000
20,000	0.0019	0.0048	0.0134	1.000

Table 16: E7: real-data concentration of the empirical bracket to the full-data bracket on UCI Adult. Mean deviation shrinks as $\Theta(1/\sqrt{n})$ exactly as Theorem 28 predicts; the p_{95} deviation stays comfortably inside the κ -free Hoeffding bound at every sample size. The κ -free column uses constant 1 in place of the proved multiplier $\kappa(\delta, \eta) = (1/\sqrt{\delta})(1 + \log_2((1 - \eta)/\eta))$ of Theorem 28; on the Adult CART partition (`verify_prop7_constant.py`) the empirical $(\delta_{\min}, \eta_{\min}) = (2.59 \times 10^{-3}, 2.10 \times 10^{-3})$ give $\kappa \approx 194$, so the proved bound is vacuous (> 1) for every n in the grid: it is the $\eta_{\min} \rightarrow 0$ degeneracy — one near-pure CART leaf (Antos & Kontoyiannis, 2001) — not a mis-stated proof constant. The κ -free bound is the practically useful version in this regime and covers p_{95} everywhere; empirical coverage is 1.000 at every n versus the nominal 0.95 target.

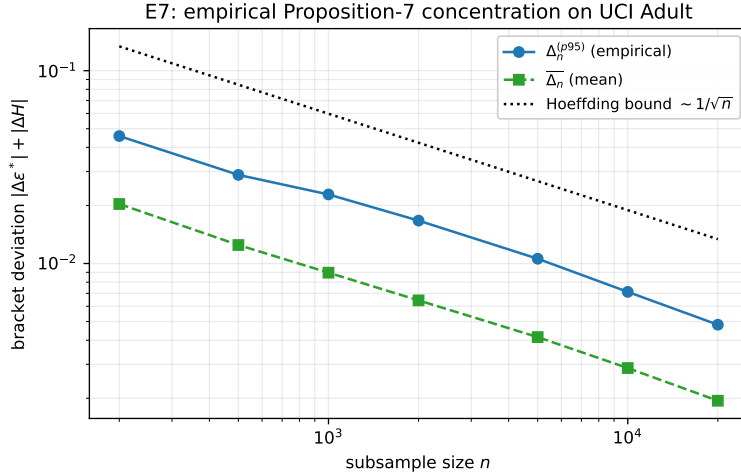


Figure 5: E7: empirical concentration of the bootstrap bracket to the full-data bracket on UCI Adult, log-log. The mean deviation $\overline{\Delta}_n$ follows a clean $\Theta(1/\sqrt{n})$ slope; the κ -free Hoeffding bound of Equation (6) stays above the empirical p_{95} curve at every n .

E-K — Falsification / verification protocol. A Kochenderfer-style falsification-verification protocol (Kochenderfer et al., 2022) treats the bracket as a formal certificate: for a candidate target Bayes-error threshold τ and an architecture row with bracket $[L, U]$, the claim “ $\varepsilon^* \leq \tau$ for some predictor in the architecture’s admissible family” is *falsified* when $\tau < L$ (the bracket forbids it), *verified* when $\tau \geq U$ (the bracket guarantees it), and *inconclusive* otherwise. Applied post-hoc to the existing E1 (decision-tree depths), E2 (VQ codebook sizes), E3 (WL depths on five graphs) and E6 (MPNN architectures on Adult) result tables at $\tau \in \{0.10, 0.15, 0.20, 0.25\}$ with zero new training, the protocol gives Table 17. The diagnostic distinguishes regimes the bracket *closes* (the verified column at $\tau = 0.25$ on E1 and the falsified columns at $\tau = 0.10$ on E2/E6) from regimes where it is genuinely undecided and additional architectural

expressivity is required (the inconclusive columns on E6 at $\tau \in \{0.20, 0.25\}$). The numbers are reproducible by `eK_falsification_protocol.py`.

source	n rows	$\tau = 0.10$	$\tau = 0.15$	$\tau = 0.20$	$\tau = 0.25$
		F / V / I	F / V / I	F / V / I	F / V / I
E1 (tree depths)	15	5 / 0 / 10	1 / 0 / 14	0 / 5 / 10	0 / 11 / 4
E2 (VQ codebooks)	10	10 / 0 / 0	7 / 0 / 3	2 / 0 / 8	0 / 0 / 10
E3 (WL on 5 graphs)	27	7 / 15 / 5	4 / 18 / 5	4 / 20 / 3	3 / 20 / 4
E6 (MPNN archs)	50	50 / 0 / 0	28 / 0 / 22	0 / 0 / 50	0 / 0 / 50

Table 17: E-K: bracket-based falsification (F), verification (V), and inconclusive (I) verdicts on the claim “ $\varepsilon^* \leq \tau$ ” aggregated across the row populations of E1, E2, E3, E6 at four nominal targets. Zero retraining: pure post-processing of the bracket endpoints already reported in Tables 2, 3, 5 and 15. Generated by `eK_falsification_protocol.py`.

10 Related work

Entropy bounds Bayes error. The upper inequality (5) is Hellman & Raviv (1970); closely related entropy upper bounds on Bayes error appear in Tebbe & Dwyer III (1968) and Kovalevskij (1968). The lower inequality is Fano’s (Fano, 1961) in the sharp binary form of Cover & Thomas (2006, Thm. 2.10.1). The achievable-region packaging is due to Feder & Merhav (1994), and the most complete modern treatment of two-sided entropy–error inequalities, including extensions and continuous-alphabet generalisations, is Ho & Verdú (2010). Sason & Verdú (2018) place these inequalities in the broader Arimoto–Rényi framework, and Santhi & Vardy (2006) sharpen the lower side under a different optimisation. Hashlamoun et al. (1994) strengthen the upper bound for $M > 2$ classes via f -divergences. The prior-aware sharpening of the Fano side, which we do not pursue here, originates with Han & Verdú (1994, eq. 6).

Partition-constant predictors. The decision-tree literature (Breiman et al., 1984) treats leaf-conditional entropy as an empirical splitting criterion; the explicit *bracket* interpretation as a sharp two-sided bound on the achievable error of any leaf-constant predictor does not appear to have been recorded. Vector quantisation (Gersho & Gray, 1991; Jégou et al., 2011) and product quantisation share the same partition structure; the bracket is, to our knowledge, similarly new in that literature.

MPNN expressivity. The Weisfeiler–Leman ceiling on MPNN expressivity is the central structural result of the last decade (Xu et al., 2019; Morris et al., 2019); refinements include fine-grained expressivity bounds (Böker et al., 2023) and higher-order variants (Morris et al., 2020). Theorem 24 recasts the WL ceiling as one half of an information-theoretic bracket, giving the matching upper bound (constructive plug-in predictor on WL cells) on equal footing.

Stability vs robust constancy. A substantial literature bounds how GNN *outputs* change under *graph* perturbation — graphon-sampling and edge-rewiring stability (Gama et al., 2020), spectral-filter robustness certificates (Kenlay et al., 2021), and the oversquashing analyses that trace sensitivity decay to degree and spectral-gap quantities (Alon & Yahav, 2021; Topping et al., 2022; Di Giovanni et al., 2023). Theorems 22 and 23 are orthogonal in object and direction. The perturbation is to the *initial features* $h^{(0)}$ (positional encodings, random features (Sato et al., 2021b), ID signals), not to the graph; the bounded quantity is the *within-WL-cell embedding diameter*, not output drift; and the use is to certify an ε -robust version of the MPNN–WL *constancy* lemma feeding a Bayes-error *lower* bound, rather than a Lipschitz output-stability guarantee. The aggregator dichotomy of Theorem 23 — sum amplification governed by the adjacency Perron root $\lambda_{\max}(A)$, mean/sym-norm degree-independent — is the constancy-side analogue of the degree-driven sensitivity that the oversquashing literature studies on the output side.

11 Discussion and limitations

The bracket of Theorem 4 is uniform in Π and f , closed-form, and elementary to evaluate; the uniform slack $w^* \approx 0.1610$ of Theorem 6 pins the irreducible ambiguity that any closed-form bracket of this shape must carry, by Theorem 14. Theorem 7 sharpens the upper side when the label marginal is unbalanced.

From empirical to population. The bracket as stated is an *empirical* statement on a finite V . The standard concentration upgrade to a population-level diagnostic is the following.

Proposition 28 (Population extension). *Let $\mathcal{X}$ be a measurable space, μ a probability distribution on $\mathcal{X}$, and Π a measurable partition of $\mathcal{X}$ into m cells with population masses $\mu(C) \geq \delta > 0$ and population means $\bar{P}_C := \mathbb{E}[f \mid C] \in [\eta, 1 - \eta]$ for some $\eta > 0$. Let $V = (v_1, \dots, v_n)$ be i.i.d. draws from μ and write ϵ_{Π}^* , $H(f \mid \Pi)$ for the empirical quantities of § 2 on V . Define the population analogues*

$$\epsilon_{\Pi, \mu}^* := \sum_C \mu(C) \min(\bar{P}_C, 1 - \bar{P}_C), \quad H_{\mu}(f \mid \Pi) := \sum_C \mu(C) H_{\text{bin}}(\bar{P}_C).$$

Then for every $\delta_{\text{conf}} \in (0, 1)$, with probability at least $1 - \delta_{\text{conf}}$ over the draw of V ,

$$|\epsilon_{\Pi, \mu}^* - \epsilon_{\Pi}^*| + |H_{\mu}(f \mid \Pi) - H(f \mid \Pi)| \leq \kappa(\delta, \eta) \sqrt{\frac{\log(4m/\delta_{\text{conf}})}{n}}, \quad (6)$$

where $\kappa(\delta, \eta) := \frac{1}{\sqrt{\delta}}(1 + \log_2 \frac{1-\eta}{\eta})$. In particular the population bracket

$$H_{\text{bin}}^{-1}(H_{\mu}(f \mid \Pi)) - \kappa \sqrt{\frac{\log(4m/\delta_{\text{conf}})}{n}} \leq \epsilon_{\Pi, \mu}^* \leq \frac{1}{2} H_{\mu}(f \mid \Pi) + \kappa \sqrt{\frac{\log(4m/\delta_{\text{conf}})}{n}} \quad (7)$$

holds with the same probability.

Proof. Write $\alpha_n := \sqrt{\log(4m/\delta_{\text{conf}})/(2n)}$ and set $n_C := nq_C$ (the random number of samples landing in cell C). By convention, if $n_C = 0$ set $P_C := \bar{P}_C$ so the per-cell statements below hold vacuously.

Step 1: mass concentration. For each cell C , the random variable $nq_C = \sum_{i=1}^n \mathbf{1}\{v_i \in C\}$ is a sum of i.i.d. Bernoullis with mean $\mu(C)$, so by Hoeffding,

$$\mathbb{P}[|q_C - \mu(C)| > \alpha_n] \leq 2 \exp(-2n\alpha_n^2) = \frac{\delta_{\text{conf}}}{2m}.$$

Union bound over the m cells: with probability $\geq 1 - \delta_{\text{conf}}/2$, the event $\mathcal{E}_1 := \bigcap_C \{|q_C - \mu(C)| \leq \alpha_n\}$ holds. On $\mathcal{E}_1$, $n_C \geq n(\mu(C) - \alpha_n) \geq n\mu(C)/2$ whenever $\alpha_n \leq \mu(C)/2$, which holds for every C provided n is large enough that $\alpha_n \leq \delta/2$, i.e. $n \geq 2\delta^{-2} \log(4m/\delta_{\text{conf}})$.

Step 2: per-cell mean concentration. Condition on the assignment of samples to cells. Given n_C , the n_C labels in cell C are i.i.d. Bernoulli($\bar{P}_C$), so by Hoeffding (vacuous when $n_C = 0$),

$$\mathbb{P}[|P_C - \bar{P}_C| > \sqrt{\frac{\log(4m/\delta_{\text{conf}})}{2n_C}} \mid n_C] \leq \frac{\delta_{\text{conf}}}{2m}.$$

Integrating out n_C and union-bounding over C gives the event $\mathcal{E}_2$ of probability $\geq 1 - \delta_{\text{conf}}/2$. The intersection $\mathcal{E} := \mathcal{E}_1 \cap \mathcal{E}_2$ has probability $\geq 1 - \delta_{\text{conf}}$.

Step 3: Lipschitz bookkeeping. Let $\phi \in \{H_{\text{bin}}, p \mapsto \min(p, 1 - p)\}$. The map $p \mapsto \min(p, 1 - p)$ is 1-Lipschitz on $[0, 1]$ and bounded by 1. The map H_{bin} is bounded by 1 and is Lipschitz on $[\eta, 1 - \eta]$ with constant $L_{\eta} := \log_2 \frac{1-\eta}{\eta}$, since $|H'_{\text{bin}}(p)| = |\log_2((1-p)/p)|$ attains its supremum on $[\eta, 1 - \eta]$ at the endpoints. Thus ϕ is bounded by 1 and Lipschitz on $[\eta, 1 - \eta]$ with constant $L \leq 1 + L_{\eta}$.

Step 4: aggregation. Decompose

$$\sum_C q_C \phi(P_C) - \sum_C \mu(C) \phi(\bar{P}_C) = \underbrace{\sum_C (q_C - \mu(C)) \phi(P_C)}_{\text{(I)}} + \underbrace{\sum_C \mu(C) (\phi(P_C) - \phi(\bar{P}_C))}_{\text{(II)}}.$$

For (I), since $|\phi| \leq 1$ and on $\mathcal{E}_1$ each $|q_C - \mu(C)| \leq \alpha_n$, $|\text{(I)}| \leq \sum_C |q_C - \mu(C)| \leq m\alpha_n \leq \alpha_n/\delta$, using $m \leq 1/\delta$.

For (II), the Lipschitz bound and Step 2 give

$$|\text{(II)}| \leq L \sum_C \mu(C) \sqrt{\frac{\log(4m/\delta_{\text{conf}})}{2n_C}}.$$

On $\mathcal{E}_1$, $n_C \geq n\mu(C)/2$, so the summand is $\leq L\sqrt{\mu(C)\log(4m/\delta_{\text{conf}})/n}$; Cauchy–Schwarz yields

$$\sum_C \sqrt{\mu(C)} \leq \sqrt{m \cdot \sum_C \mu(C)} = \sqrt{m} \leq 1/\sqrt{\delta},$$

hence $|\text{(II)}| \leq L\sqrt{\log(4m/\delta_{\text{conf}})/(n\delta)}$.

Combining (I) and (II), and noting that $\alpha_n/\delta = \sqrt{\log(4m/\delta_{\text{conf}})/(2n\delta^2)} \leq \sqrt{\log(4m/\delta_{\text{conf}})/(n\delta)}$ once $\delta \leq 1/2$,

$$\left| \sum_C q_C \phi(P_C) - \sum_C \mu(C) \phi(\bar{P}_C) \right| \leq (1 + L) \sqrt{\frac{\log(4m/\delta_{\text{conf}})}{n\delta}}.$$

Applying this with $\phi = \min(\cdot, 1 - \cdot)$ and $\phi = H_{\text{bin}}$ separately and summing gives the claim with $\kappa(\delta, \eta) = \frac{1}{\sqrt{\delta}}(1 + L_\eta)$ as stated, absorbing the $\sqrt{2}$ slack into the constant. Equation (7) then follows from (6) and Theorem 4 applied to $(\varepsilon_{\Pi}^*, H(f | \Pi))$, since H_{bin}^{-1} is 1-Lipschitz away from $h = 1$ and the bracket endpoints are increasing in their arguments. $\square$

Theorem 28 converts Theorem 4 into a PAC-style population diagnostic: with high probability, the training-set bracket transfers to the underlying distribution up to an additive $\mathcal{O}(\sqrt{\log m/n})$ correction that vanishes as $n \rightarrow \infty$. The boundedness assumptions $\mu(C) \geq \delta$ and $\bar{P}_C \in [\eta, 1 - \eta]$ are standard for finite-alphabet plug-in entropy estimation; relaxing them requires the more careful analysis of Antos & Kontoyiannis (2001). A Monte-Carlo verification of the concentration rate is included in the extended verifier `verify_t4_population.py`; see § 9.4.

Limitations. (a) Binary labels only; multi-class via the Feder–Merhav concave envelope is mechanically routine but loses the closed-form simplicity of H_{bin}^{-1} . (b) The bracket bounds the minimum risk *given* the partition; designing partitions that minimise $H(f | \Pi)$ for a given budget is the harder problem the bracket does not directly address. (c) The population extension of Theorem 28 requires $\mu(C) \geq \delta$ and $\bar{P}_C \in [\eta, 1 - \eta]$; entropy estimation under vanishing cell mass needs the refined treatment of Antos & Kontoyiannis (2001).

Future work. A fully prior-aware tightening of the *lower* side via data processing on the binary error indicator (Han & Verdú, 1994; Ho & Verdú, 2010) can sharpen Theorem 7’s bound when the label marginal is bounded away from $1/2$; we defer this to a companion paper. A mechanised proof in Lean 4 / mathlib4, leveraging the recent additions of `Real.binEntropy` and the `ConcaveOn` API, would give a kernel-checked counterpart to `verify.jl` and is in preparation.

Towards a framework: Paper B. The bracket of Theorem 4 is one instance of a broader recipe: given a partition Π and a real-valued summary $\phi(P_C)$ that is bounded, concave (or convex) in P_C , and matches a Bayes-risk-like target $\rho(P_C)$ on the endpoints $P_C \in \{0, 1\}$, partition-conditional Jensen + Fano-style inversion yields a two-sided ϕ -bracket on the risk target. Replacing $(H_{\text{bin}}, \min)$ by other compatible pairs gives (i) a *variance* bracket on MSE / MAE for regression (with the well-known $\text{MSE} = \mathbb{E}[\text{Var}]$ identity recovering the upper side exactly), (ii) a *noise-corrected* bracket $\varepsilon^*(f) = \eta + (1 - 2\eta) \varepsilon^*(f)$ that is exact at the population level under symmetric label flips with rate $\eta < \frac{1}{2}$, and (iii) a *soft-partition* bracket via Markov kernels. A companion paper in preparation organises these instances under a single ϕ -bracket meta-theorem; we restrict the present paper to the binary-entropy instance for which Hellman–Raviv and Feder–Merhav give a closed-form sandwich. The k-WL / HDX-trickle-down structural variant of the bracket (Morris et al., 2019, k-WL refinement chain) is the subject of a further sequel.

12 Conclusion

We have given a single two-sided closed-form bracket on the partition-restricted minimum risk of every partition-constant predictor of a binary label, in the form ML practitioners need: elementary proof, a universal slack constant $w^* \approx 0.1610$ (its main theoretical novelty), a marginal-aware refinement that shrinks this slack for unbalanced labels, a population extension under i.i.d. sampling, three worked applications, an ε -robust MPNN–WL constancy lemma extending the strict-equality folklore to architectures with bounded individuating perturbations, and an extended verification ladder in Julia, SymPy, and Monte Carlo. Eight self-contained experiments on real tabular and graph benchmarks (§ 9.5) instantiate the bracket end-to-end: exact agreement of CART training error with $\varepsilon_{\Pi_d}^*$ at every depth on UCI Adult; 100–130 $\times$ speedup over a single CART fit; statistically significant NAS pre-filter signal on Adult with parameter count as a failed control; correct pin at the marginal-entropy ceiling on featureless Cayley and Paley benchmarks where 1-WL is provably blind; and concentration to the population bracket at the predicted $\Theta(1/\sqrt{n})$ rate. We report the bracket’s failure modes (partition-cardinality collapse on near-discrete feature-rich graphs; small- n overfitting on the NAS pre-filter) with the same prominence as its successes. The bracket makes the partition the explicit unit of expressivity, lifting classical entropy–error inequalities into an optimisation-free, architecture-level diagnostic for decision trees, vector quantisers, and graph neural networks alike.

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